WETLAND RESTORATION PLAN FOR MAKINDYE SSABAGABO MUNICIPAL 2025-2030



Approved by the Council under
Minute Number MSMC/COU/03/26/06/25

Executive Summary

Makindye Ssabagabo Municipality, located within the rapidly urbanizing periphery of Kampala, is home to several ecologically and socioeconomically vital wetland systems, including Kaliddubi, Kyetinda, and Lufuka wetlands. These wetlands play a critical role in flood control, water purification, climate regulation, and community livelihoods. However, due to escalating urban encroachment, pollution, poor land use, and institutional weaknesses, nearly 60% of the wetlands in the municipality are currently degraded—a significant increase from 30% in 2005.

In response to this alarming degradation, the Wetland Restoration and Management Plan has been developed as a comprehensive and strategic framework aimed at reversing wetland loss and promoting sustainable use. The plan is anchored in national laws and policies, such as the National Environment Act (2019) and the National Wetlands Policy, and aligns with international conventions, including the Ramsar Convention and the Sustainable Development Goals (SDGs).

Purpose and Goal
The purpose of the plan is to provide a coordinated and community-led roadmap to restore degraded wetlands, enforce legal protections, and integrate wetland management into broader municipal planning. Its goal is to restore ecological integrity, sustain livelihoods, and enhance climate resilience through inclusive and evidence-based interventions.

Vision and Mission

Vision

"A municipality where wetlands are fully restored, sustainably managed, and valued as vital ecosystems that support biodiversity, climate resilience, and livelihoods".

Mission: To mobilize local communities, institutions, and stakeholders in implementing effective restoration and protection measures through inclusive planning, regulation, education, and eco-friendly livelihoods.

Strategic Result Areas and Key Interventions
The plan is structured around six strategic Result Areas:

1. **Water Resources Management** – Rehabilitating stormwater systems, controlling pollution, and restoring natural water flow, particularly in wetlands like Lubigi and Kyetinda.
2. **Service Providers** – Strengthening capacities of nursery operators and restoration input suppliers through training, equipment support, and registry development.
3. **Community Engagement and Behavior Change** – Mobilizing households, promoting awareness, forming wetland user groups, and incentivizing eco-friendly practices.
4. **Wetland Governance** – Establishing bylaws, demarcating wetland zones using GIS tools, and enhancing interagency collaboration through taskforces and data sharing.
5. **Livelihood Strengthening** – Introducing sustainable income sources (e.g., aquaculture, beekeeping, eco-tourism), creating green jobs, and supporting eco-enterprise development.
6. **Wetland Monitoring and Assessment** – Conducting mapping, installing real-time monitoring systems, and establishing frameworks for restoration tracking and adaptive management.

Implementation Framework
The implementation timeline spans 2026 to 2030, with actions prioritized by urgency and feasibility. Institutional responsibility lies with Makindye Ssabagabo Municipal Council (MSMC), supported by national agencies (e.g., NEMA, MWE), NGOs, and community structures. The plan includes short-, medium-, and long-term actions, ensuring flexibility and scalability.

Budget and Resource Mobilization
The total estimated cost of implementing the plan is UGX 70 billion, covering all six result areas. Funding will be mobilized from government allocations, development partners (e.g., UNDP, World Bank), NGO partnerships, and private sector investments. Community contributions, especially in labor and surveillance, are also anticipated.

Stakeholder Engagement and Governance
A multi-level stakeholder structure is proposed:

A Wetland Management Taskforce chaired by the Town Clerk will oversee coordination. Division-level environmental committees and community wetland monitors will ensure local action and feedback. Roles of key actors—including MSMC, NEMA, MWE, CSOs, religious leaders, and private sector—are clearly delineated to ensure accountability and collaboration.

Monitoring, Evaluation, and Learning (MEL)
A robust MEL framework defines indicators, baselines, and annual targets for each result area. Methods include GIS mapping, knowledge-attitude-practice (KAP) surveys, and quarterly performance reviews. The plan adopts a learning-by-doing approach, allowing for adaptive management and stakeholder feedback integration.

Conclusion
The Makindye Ssabagabo Wetland Restoration and Management Plan presents a transformative opportunity to reverse wetland degradation and position the municipality as a leader in urban ecological stewardship. By aligning policy, science, community action, and sustainable development, the plan aims to restore over 640 hectares of wetland, improve local climate resilience, and uplift wetland-dependent households—securing the environmental future of the municipality.

Forward

Wetlands are among the most valuable ecosystems in our municipality, providing flood control, water purification, biodiversity habitat, and vital livelihoods for many of our residents. Yet, over the past two decades, these natural assets have come under unprecedented pressure from rapid urbanization, unplanned settlements, pollution, and unsustainable land use. The result has been significant degradation, threatening not only the ecological health of our wetlands but also the well-being of our communities.

The Makindye Ssabagabo Wetland Restoration and Management Plan is a timely and strategic response to this challenge. Developed through scientific assessment, stakeholder consultations, and alignment with both national and international environmental frameworks, this plan offers a clear roadmap for reversing wetland loss and ensuring their sustainable management. It emphasizes community participation, legal enforcement, livelihood diversification, and the integration of wetland conservation into broader municipal planning.

Successful implementation of this plan will require the commitment of all stakeholders, including government agencies, community leaders, civil society, the private sector, and residents. Together, we can restore our wetlands to full health, strengthen our resilience to climate change, and secure the ecological and socio-economic benefits they provide for current and future generations.

On behalf of Makindye Ssabagabo Municipal Council, I commend all those who contributed their time, expertise, and resources to the development of this plan. Let us now work collectively to turn its strategies into tangible results.



Ssemwanga Godfrey
His Worship, Mayor
Makindye Ssabagabo Municipality

Acknowledgment

The preparation of the Makindye Ssabagabo Wetland Restoration and Management Plan would not have been possible without the valuable contributions and commitment of many individuals and institutions.

We extend our sincere appreciation to the World Bank for its financial and technical support through the Greater Kampala Metropolitan Area Urban Development Program (GKMA-UDP), which facilitated the development of this plan.

Special thanks go to the Political Leadership and Technical Staff of Makindye Ssabagabo Municipal Council, whose dedication to environmental stewardship has been instrumental throughout the planning process. We are grateful to the Division Leaders, Local Council representatives, and Community Members who participated in consultations, shared their experiences, and provided invaluable local knowledge to inform this plan.

Finally, we recognize the technical expertise provided by Environment and Earth Sciences Consult Limited, whose team conducted the ecological assessments, stakeholder engagements, and documentation of the plan.

We hope that the collective effort and commitment demonstrated during the preparation of this plan will continue into its implementation, ensuring the restoration, protection, and sustainable management of wetlands for the benefit of current and future generations.



Richard K Monday
Town Clerk
Makindye Ssabagabo Municipality

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Acronyms

GCF	Green Climate Fund
NEMA	National Environment Management Authority
UNDP	United Nations Development Program
GPS	Global Positioning System
MEO	Municipal Environmental Officer
MWE	Ministry of Water and Environment
EWS	Early Warning Systems

Chapter One: Introduction

1.1 Introduction

Wetlands play a crucial role in supporting biodiversity, regulating water cycles, and providing livelihoods for communities. In Makindye Ssabagabo Municipality, wetlands serve as critical natural assets supporting both ecological balance and local livelihoods. However, rapid urbanization, land encroachment, pollution, and unsustainable land-use practices have led to significant degradation of these wetlands, threatening their ecological integrity and the benefits they provide.

Recognizing the urgent need for intervention, this Wetland Restoration and Management Plan outlines strategic approaches to rehabilitate, protect, and sustainably manage wetlands within the municipality. The plan integrates scientific assessments, policy guidelines, community involvement, and institutional arrangements to ensure effective restoration and long-term conservation. Uganda's National Wetlands Policy is guided by the National Environment Act (2024) and global frameworks, including the Ramsar Convention and the UN Sustainable Development Goals (SDGs).

1.2 Objectives

The objectives of the plan are;

1. To restore and rehabilitate degraded wetlands and enhance their ecological functions, especially in critical areas such as Kaliddubi, Kyetinda, and Lufuka that have experienced significant loss of vegetation and water retention capacity.
2. To protect wetlands through legal demarcation, zoning, and enforcement of bylaws, in collaboration with NEMA and local councils, to curb illegal settlements, sand mining, and unsanctioned agricultural expansion.
3. To increase public awareness and change perceptions toward wetlands by implementing targeted education campaigns and engaging schools, religious institutions, and local leaders to promote wetland stewardship.
4. To provide sustainable and eco-friendly livelihood alternatives to wetland-dependent households, particularly those involved in bricklaying, informal farming, and sand mining, thereby reducing human pressure on fragile wetland ecosystems.
5. To strengthen institutional capacity at the municipal and division levels for wetland monitoring, policy enforcement, and integrated environmental planning, including the establishment of monitoring systems and training of environmental officers.
6. To foster community participation and stakeholder collaboration in wetland management through the formation of local environment committees, wetland user groups, and partnerships with civil society and academic institutions.
7. To integrate wetland restoration into broader municipal development plans, ensuring that infrastructure expansion and urban growth are aligned with environmental sustainability principles.

1.3 Purpose of this plan

The purpose of the Plan is to provide a strategic, coordinated, and community-driven framework for the restoration, protection, and sustainable management of wetlands within Makindye Ssabagabo Municipality in order to enhance ecosystem integrity, mitigate climate risks, and support inclusive socio-economic development.

Re-establishment of hydrological regimes and associated ecohydrological connectivity: The primary purpose often centers on restoring the characteristic wetland hydrology, which is the foundational driver of wetland ecosystem structure and function. This involves engineering interventions to re-establish natural water flow pathways (e.g., re-meandering channelized streams, removing artificial impoundments), restoring historical hydroperiods (duration and frequency of inundation/saturation), and optimizing water budgets (precipitation, evapotranspiration, surface inflow/outflow, groundwater interaction). The aim is to reinstate the dynamic interplay between surface water and groundwater that defines the wetland's unique physicochemical environment, thereby promoting the development of characteristic hydric soil conditions and the recruitment and persistence of obligate and facultative wetland vegetation. This re-establishment of ecohydrological connectivity ensures the appropriate water availability and flow dynamics essential for the target wetland ecosystem.

Enhancement of Biodiversity and Ecological Community Structure: A critical purpose is to foster the recovery of native biodiversity, including characteristic flora and fauna, and to restore the integrity of the ecological community structure. This involves implementing strategies to facilitate the colonization and establishment of desired native plant species (e.g., through seed bank activation, direct planting of native propagules, control of invasive exotic species) and to create suitable habitats for wetland-dependent wildlife (e.g., through microtopographic manipulations, creation of diverse vegetation mosaics, provision of nesting/foraging sites). The objective is to increase species richness and abundance, particularly of threatened or endangered wetland species, and to re-establish trophic interactions and ecological processes (e.g., nutrient cycling, decomposition) characteristic of a healthy, functioning wetland ecosystem.

Restoration of Ecosystem Services and Functional Capacity: Wetland restoration plans aim to reinstate and enhance the provision of vital ecosystem services that have been degraded or lost due to anthropogenic impacts. This encompasses a range of functions such as water quality improvement (e.g., nutrient retention and denitrification, sediment sequestration, contaminant removal), flood attenuation and stormwater management (e.g., water storage, peak flow reduction), groundwater recharge, carbon sequestration, and shoreline stabilization. The technical approach involves designing the restored wetland to optimize its capacity for these specific functions, often through considerations of wetland morphology, vegetation type and density, and hydraulic residence time, thereby maximizing the socio-economic and ecological benefits derived from the wetland.

Mitigation of Anthropogenic Disturbances and Enhancement of System Resilience: A forward-looking purpose involves addressing the root causes of degradation and building the long-term resilience of the wetland ecosystem against future anthropogenic and climatic disturbances. This includes implementing measures to abate ongoing sources of pollution (e.g., agricultural runoff, industrial discharge), managing altered land-use patterns in the surrounding watershed, and developing adaptive management strategies to account for the impacts of climate change (e.g., sea-level rise, altered precipitation patterns). The goal is to create a robust and self-sustaining wetland system that can better withstand environmental stressors, maintain its ecological integrity, and continue to provide essential services in the face of evolving environmental conditions.

1.4Vision, goal, and mission

Vision

"A municipality where wetlands are fully restored, sustainably managed, and valued as vital ecosystems that support biodiversity, climate resilience, and livelihoods."

Goal

To restore degraded wetlands, prevent further encroachment, and promote sustainable use of wetland resources for the benefit of current and future generations in Makindye Ssabagabo Municipality.

Mission

To mobilize local communities, institutions, and stakeholders in the implementation of effective wetland restoration and protection measures through inclusive planning, regulatory enforcement, education, and environmentally friendly livelihoods.

1.5 Wetland Restoration Process

In Uganda, wetland restoration follows a structured approach guided by national policies and international frameworks. It involves the following steps.

a) Assessment & Planning—This involves identifying degraded wetlands, assessing the extent of damage, and developing restoration plans. Uganda's National Wetlands Policy (1995) emphasizes sustainable wetland management and conservation.
b) Community Engagement—Local communities are involved in restoration efforts through awareness campaigns and alternative livelihood programs to reduce dependence on wetland resources.
c) Rehabilitation Measures—These include reforestation, removal of invasive species, and reintroducing native vegetation to restore ecological balance.
d) Policy & Legal Enforcement—Uganda enforces wetland protection through the National Environment Act (2019) and other regulations that prohibit encroachment and promote sustainable use.
e) Monitoring & Evaluation—Continuous assessment ensures restoration efforts are effective, and wetlands remain protected

Chapter Two: Situational Analysis

2.1 Introduction

This chapter presents a comprehensive situational analysis of wetlands within the Makindye Ssabagabo municipality, examining their coverage, current state, strengths and weaknesses, management approaches, and the policy and legal frameworks governing their protection. Additionally, it explores institutional arrangements that influence wetland conservation efforts. Understanding these factors is crucial for the establishment of a foundation for effective restoration and sustainable management strategies that align with environmental conservation goals and community needs.

2.2 Coverage and benefits of wetlands

The wetlands in Makindye Ssabagabo Municipality cover a total area of 1,600.8 hectares (18.9% of the municipality's surface area). Ndejje division (58%) has the biggest coverage of wetlands in the municipality, followed by Masajja division (32%) and then Bunamwaya division (10%) (**Figure 1**).

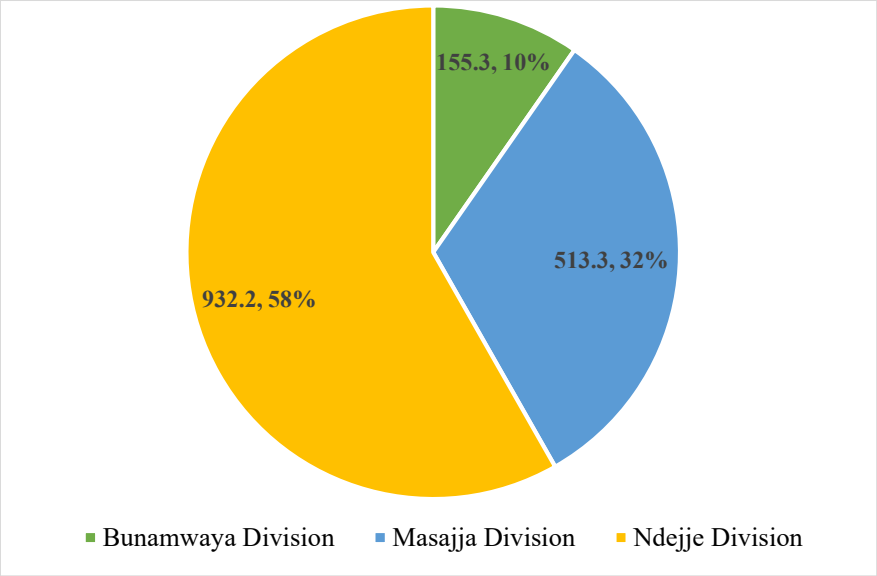


Figure 1: Coverage of wetlands by division in Makindye Ssabagabo Municipality

The major wetland systems in the municipality include Kaliddubi, Kyetinda, Lufuka, and Mayanja-Lufuka (**Figure 2**).

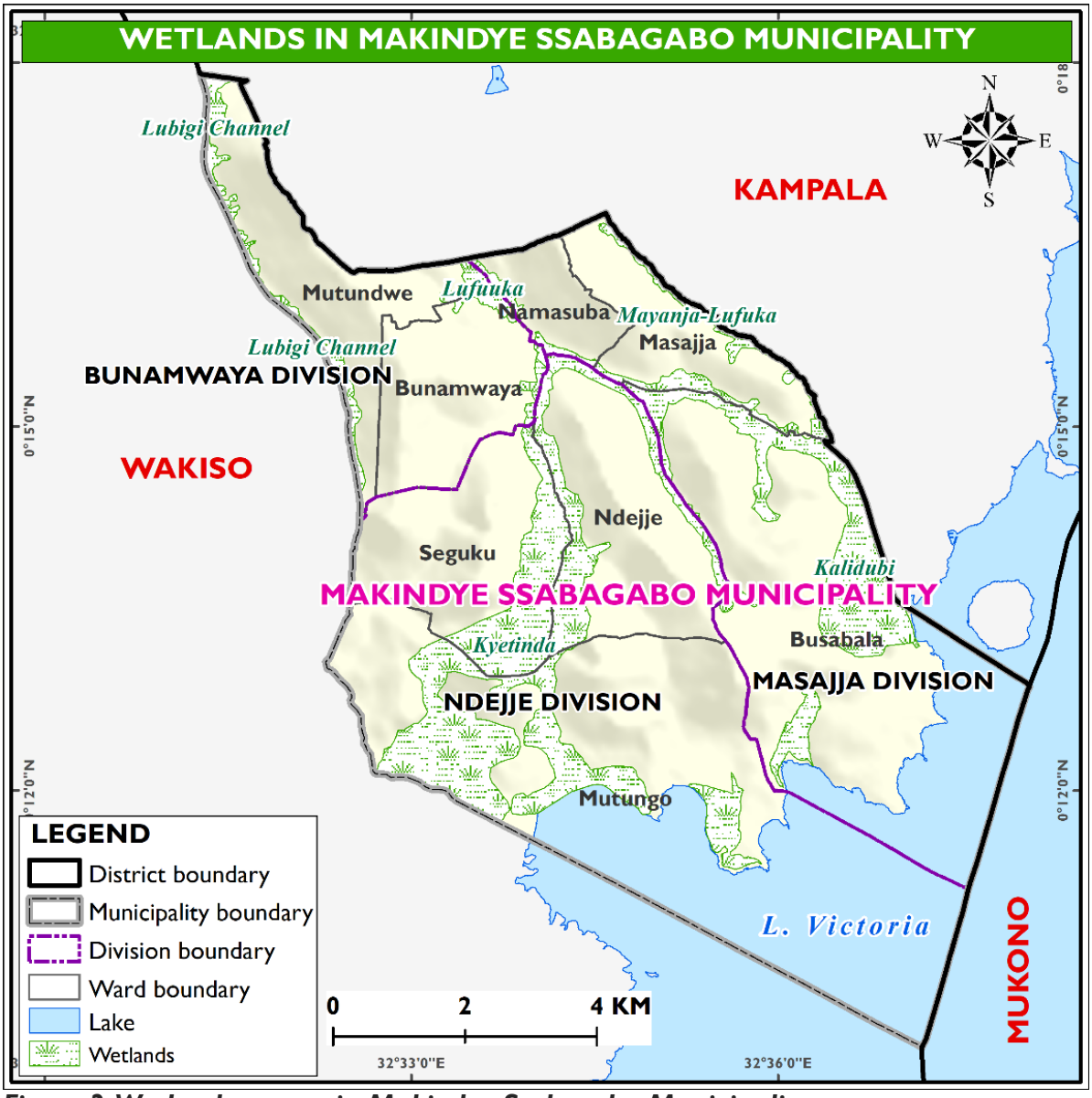


Figure 2: Wetland systems in Makindye Ssabagabo Municipality

The coverage of intact wetlands in 2005 was 1,118.7 ha (13.2% of the municipality's surface area) and 692.7 ha (8.2% of the municipality's surface area) in 2015. In 2025, however, the coverage of intact wetlands is 640.9 ha (7.5% of the municipality's surface area) (**Figure 3**). Between 2005 and 2025, Makindye Ssabagabo Municipality lost 477.8 ha of wetlands (5.6% of the municipality's surface area). The decrease in wetland coverage can be attributed to increasing encroachment by farmlands and built-up areas.

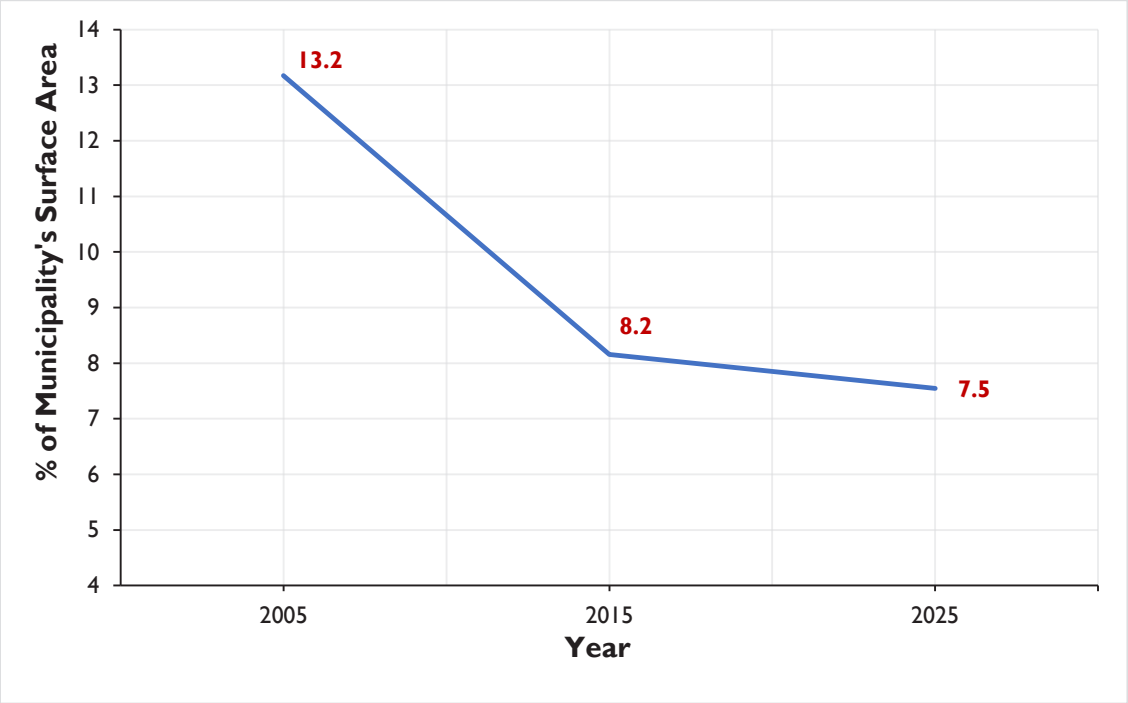


Figure 3: Coverage of intact wetlands in Makindye Ssabagabo Municipality in 2005, 2015, and 2025

Seasonality: About 69.5% of wetlands (1,112.4 Ha) are permanent, whereas 30.5% of wetlands (488.4 Ha) are seasonal (**Figure 4**).

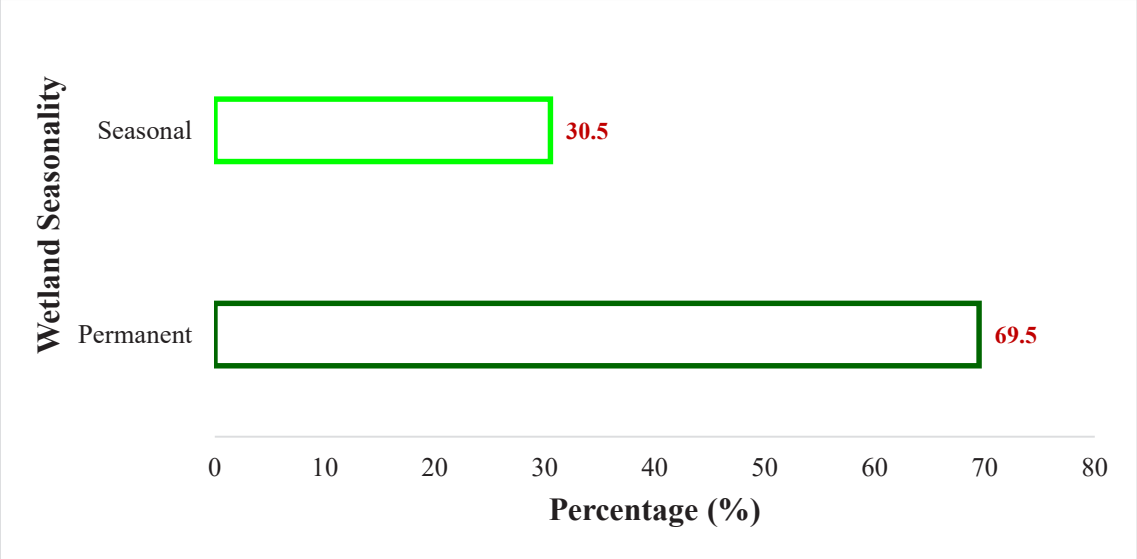


Figure 4: Wetland seasonality in Makindye Ssabagabo Municipality

Wetland ecosystem services: The communities revealed that the major benefits derived from the wetlands of Makindye Ssabagabo Municipality include clay mining (11.7%), sand mining (11%), climate regulation (10.2%), collection of herbal medicine (9.1%), grazing (8.2%), collection of craft materials (7.7%), grasses for mulching (6.4%), water for irrigation (5.5%), water for domestic use (5.5%), and flood regulation (4.2%) (**Figure 5**). However, some of these benefits or activities, such as crop cultivation and sand and clay mining, contribute to the degradation of wetlands, and the municipality should regulate them.

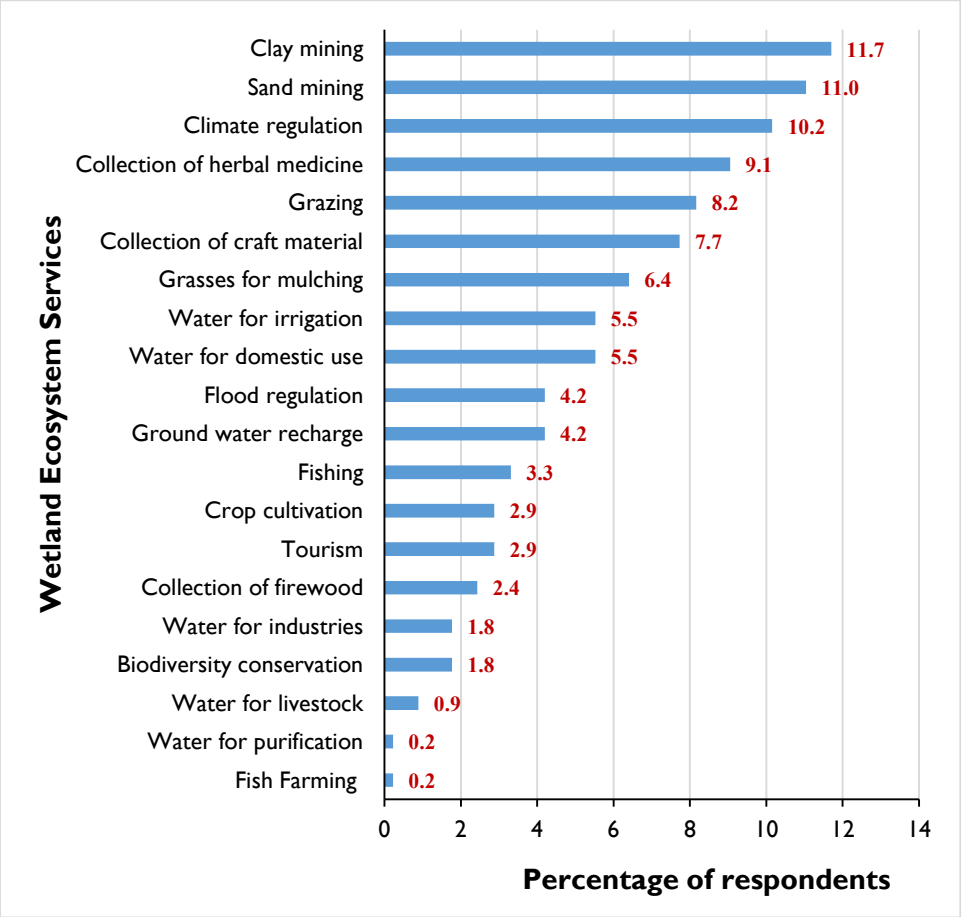


Figure 5: Wetland ecosystem services in Makindye Ssabagabo Municipality

2.3 State of wetlands

2.3.1 Physical state

2.3.1.1 Wetland classes/changes

The wetlands in Makindye Ssabagabo Municipality are composed of built-up areas, croplands, grasslands, open water, papyrus, and woodlands. Currently, the wetlands are dominated by land use classes, i.e., built-up areas (45%) and croplands (15%) (**Figure 7**). This shows the huge impact of urbanization and population growth on the municipality's wetlands. Open water (4.3%) is the least dominant class in the wetlands of Makindye Ssabagabo Municipality. Refer to Appendix 2 for detailed wetland inventories.

A. 2005

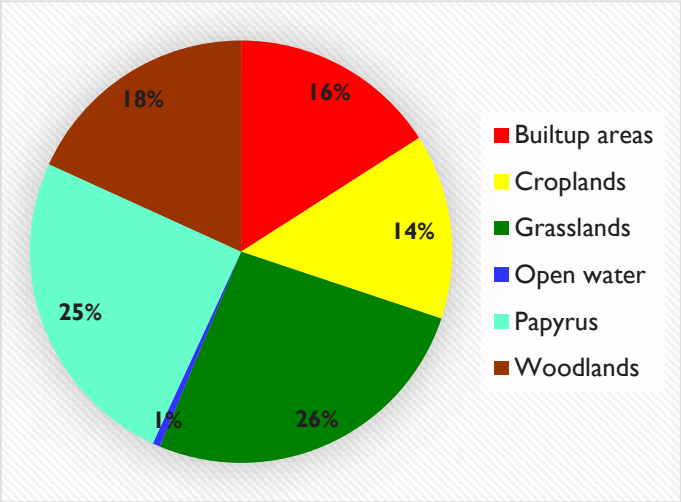


Figure 6: Wetland use/cover types in 2005

B. 2025

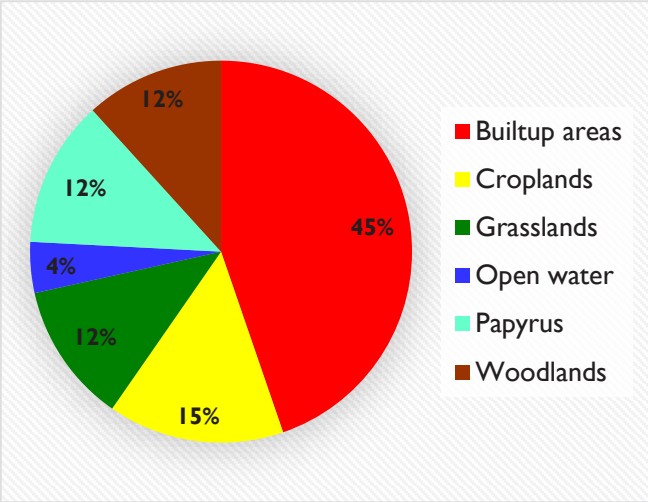
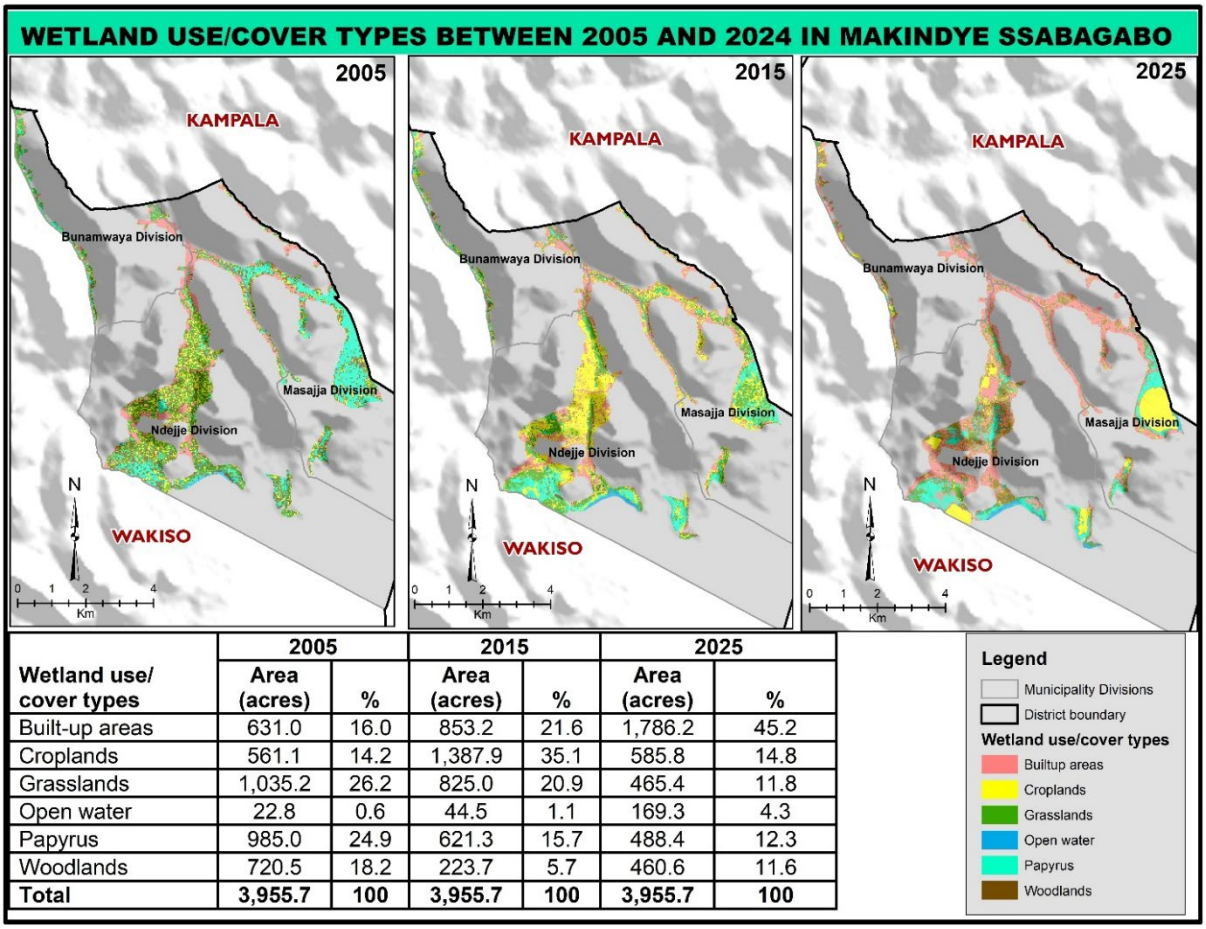


Figure 7: Wetland use/cover types in 2025

Between 2005 and 2025, built-up areas, croplands, and open water increased by 28.8%, 0.7%, and 3.7%, respectively, whereas grasslands, papyrus, and woodlands decreased by 14.3%, 12.5%, and 6.5%, respectively. The increase in open water can be attributed to drainage for agricultural expansion and diversion of water for irrigation. The increase in built-up areas and croplands can be attributed to population growth, creating demand for land for agriculture and settlement. Between 2015 and 2025, the population in Makindye Ssabagabo municipality is projected to have increased by 256,100 people (88.7%) (UBOS, 2025). The increase in built-up areas can also be attributed to the strategic placement of industries near wetlands, facilitating the direct discharge of untreated or partially treated effluents into these ecosystems (Angiro et al., 2020). The reduction in papyrus, woodlands, and grasslands can be primarily attributed to conversions to wetland use types, notably croplands and built-up areas.

The satellite images show an increase in built-up areas between 2015 and 2025 in all the wetlands of the municipality, especially in Masajja and Ndejje divisions (**Figure 8**). Pockets of croplands were also seen in 2025 at the edges of the wetlands. There is also a new trend of cultivating in the middle of major wetland systems, where visibility is hard for the municipal officers. This is common in the Kaliddubi wetland system near Lake Victoria in Busabala ward and the Kyetinda wetland system in Mutungo ward.



Wetland use/ cover types	2005		2015		2025	
	Area (acres)	%	Area (acres)	%	Area (acres)	%
Built-up areas	631.0	16.0	853.2	21.6	1,786.2	45.2
Croplands	561.1	14.2	1,387.9	35.1	585.8	14.8
Grasslands	1,035.2	26.2	825.0	20.9	465.4	11.8
Open water	22.8	0.6	44.5	1.1	169.3	4.3
Papyrus	985.0	24.9	621.3	15.7	488.4	12.3
Woodlands	720.5	18.2	223.7	5.7	460.6	11.6
Total	**3,955.7**	**100**	**3,955.7**	**100**	**3,955.7**	**100**

Figure 8: Wetland use/cover changes between 2005 and 2025 in Makindye Ssabagabo Municipality

2.3.1.2 Wetland degradation

Wetland degradation is the process by which wetland area, structure, and function have degenerated and, at times, even disappeared under the effect of natural and human activities. The wetlands report (2019) highlights anthropogenic disturbance, wetland changes, and changes in wetland productivity as indicators of wetland degradation. The wetland use/cover types were used to categorize the status of wetlands in Makindye Ssabagabo Municipality. Built-up areas, farmlands, and tree plantations were classified as degraded wetlands, while grasslands, papyrus, open water, and woodlands were classified as intact wetlands.

In the municipality, 59.7% of the wetlands (947.6 ha) are degraded, whereas 40.3% of the wetlands (640.9 ha) are intact (**Figure 9**). In all the divisions, the proportion of degraded wetlands is more than the intact wetlands, calling for urgent action to reverse the trend. Bunamwaya and Masajja divisions have the biggest percentage of degraded wetlands, each having 68.8%. Ndejje division (46.9%) has the biggest proportion of intact wetlands in Makindye Ssabagabo municipality.

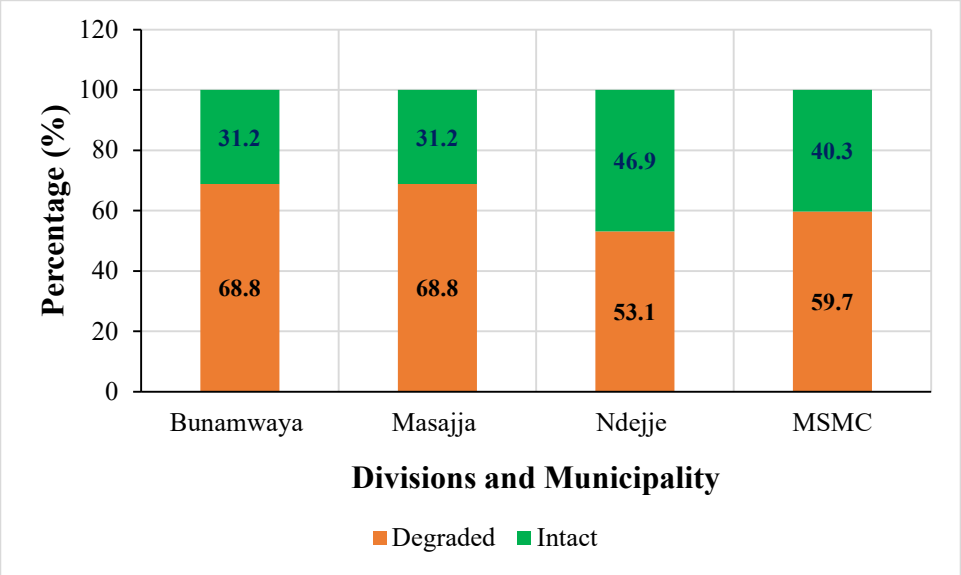


Figure 9: Status of wetlands in Makindye Ssabagabo Municipality in 2025

The trend analysis showed an increase in degraded wetlands in the municipality from 30.1% in 2005 to 59.7% in 2025 (**Figure 10**). This 29.5% increment in degraded wetlands calls for urgent action to reverse the trend.

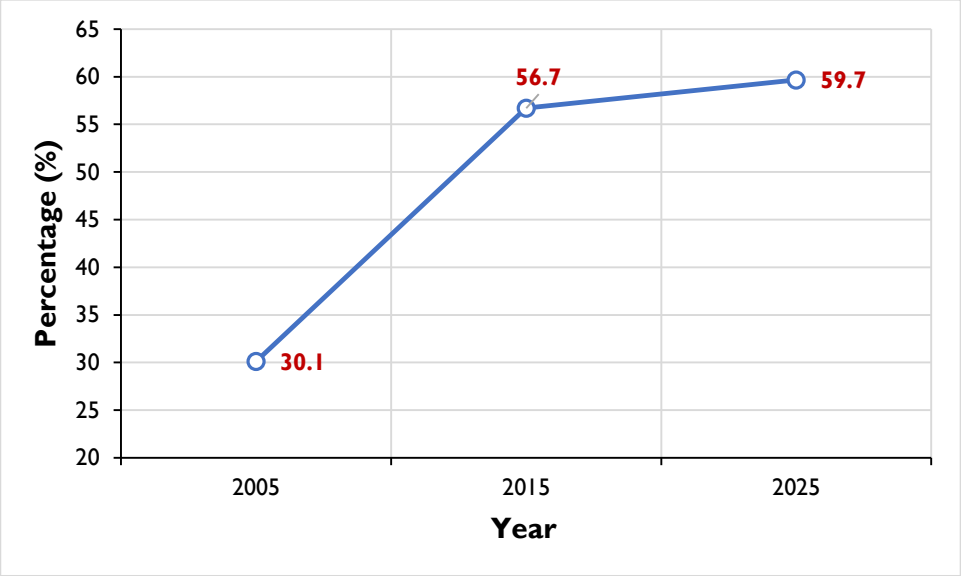


Figure 10: Percentage of degraded wetlands in Makindye Ssabagabo Municipality in 2005, 2015, and 2025

At the ward level, Bunamwaya (81.4%), Namasuba (80.6%), Masajja (78.4%), and Busabala (65.8%) have the biggest percentages of degraded wetlands (**Figure 11**). On the other hand, Mutungo (51.4%), Seguku (46.1%), Ndejje (36.9%), and Mutundwe (35.3%) have the biggest percentages of intact wetlands in the municipality.

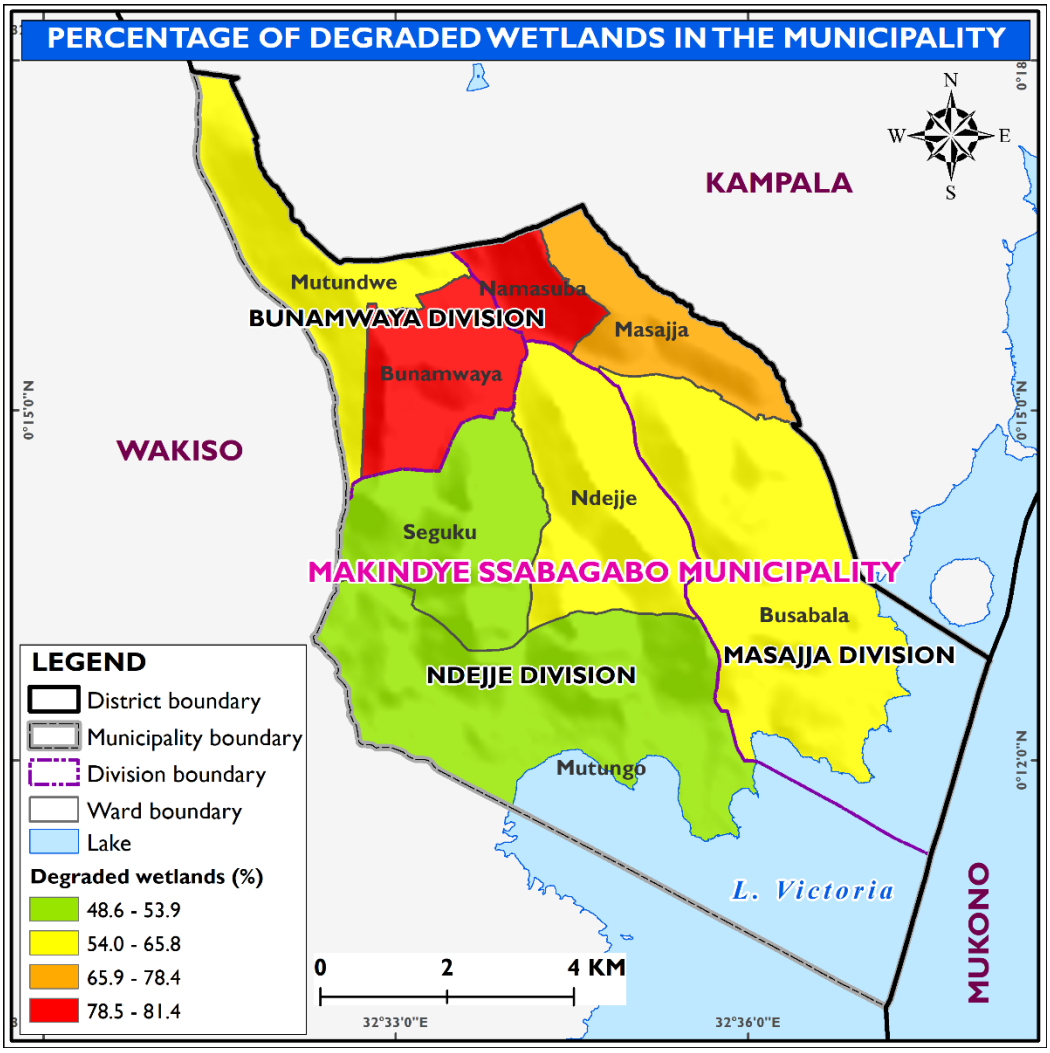
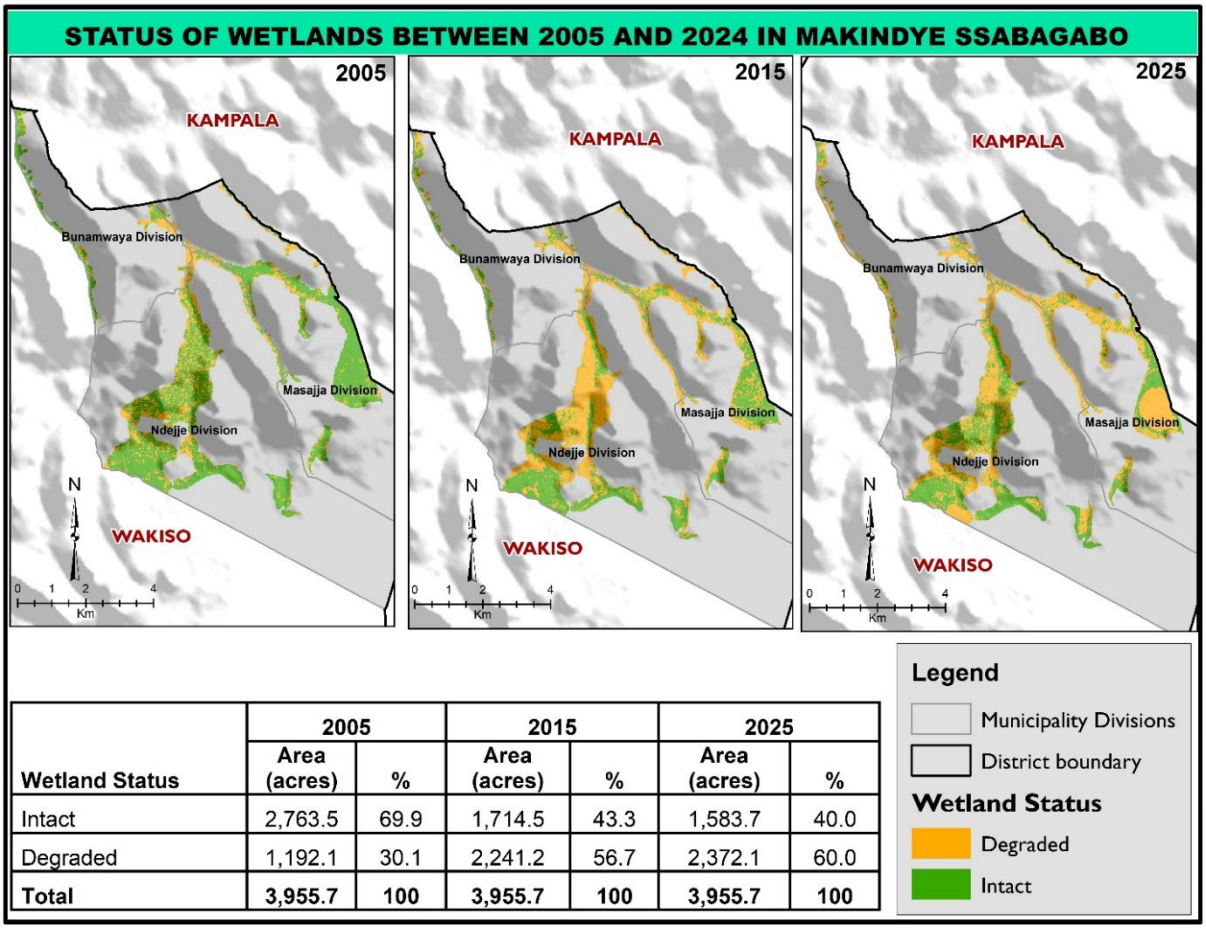


Figure 11: Percentage of degraded wetlands in the municipality in 2025

The status of wetlands in Makindye Ssabagabo municipality between 2005 and 2025 is shown in **Figure 12**.



Wetland Status	2005		2015		2025	
	Area (acres)	%	Area (acres)	%	Area (acres)	%
Intact	2,763.5	69.9	1,714.5	43.3	1,583.7	40.0
Degraded	1,192.1	30.1	2,241.2	56.7	2,372.1	60.0
Total	**3,955.7**	**100**	**3,955.7**	**100**	**3,955.7**	**100**

Figure 12: Status of wetlands in Makindye Ssabagabo municipality

2.3.1.3 Drivers of Wetland degradation

Generally, household respondents identified infrastructural development as the leading driver of wetland degradation in Makindye Ssabagabo Municipality (18.3%), with Bunamwaya division accounting for the largest share (49.2%) (**Table 1**). Other major drivers of wetland degradation identified by households include dumping of solid waste in wetlands (10.2%), brick making (9.9%), sand and clay mining (8.8%), and sand and clay mining (6.4%). Focused group discussants and key informants also highlighted urban encroachment, agricultural expansion, sand mining, the absence of bylaws on wetland management, and pollution from industrial and domestic activities as major drivers of wetland degradation. These findings underscore the urgent need for sustainable wetland management and restoration to protect these vital ecosystems.

Table 1: Drivers of wetland degradation in Makindye Ssabagabo Municipality

Drivers of wetland degradation	Bunamwaya		Masajja		Ndejje		Grand Total	
	N	%	N	%	N	%	N	%
Infrastructural development	96	49.2	22	4.4	40	23.5	158	18.3
Dumping of solid waste in wetlands	25	12.8	41	8.2	22	12.9	88	10.2
Brick making	9	4.6	51	10.3	25	14.7	85	9.9
Cultivation from wetlands	32	16.4	29	5.8	15	8.8	76	8.8
Sand and clay mining	1	0.5	38	7.6	16	9.4	55	6.4
Construction of roads and valley dams	1	0.5	42	8.5	10	5.9	53	6.1
Channeling of water from the wetland	3	1.5	41	8.2	6	3.5	50	5.8
Grazing from wetlands	2	1.0	47	9.5	1	0.6	50	5.8
Limited awareness	6	3.1	41	8.2	2	1.2	49	5.7
Burning of wetland vegetation	1	0.5	36	7.2	5	2.9	42	4.9
Establishment of washing bays	15	7.7	18	3.6	5	2.9	38	4.4
Tree planting	1	0.5	37	7.4	2	1.2	40	4.6
Excessive water abstraction	0	0.0	23	4.6	2	1.2	25	2.9
Over-harvesting of wetland benefits	3	1.5	19	3.8	3	1.8	25	2.9
Settlement	0	0.0	0	0.0	15	8.8	15	1.7
Soil erosion	0	0.0	12	2.4	1	0.6	13	1.5
Total	**195**	**100**	**497**	**100**	**170**	**100**	**862**	**100.0**



Plate 1: Waste dumped in Kaliddubi wetland, Ndejje ward, Ndejje division



Plate 2: Brick making in Kaliddubi wetland, Busabala ward, Masajja division

2.3.2 Biological State

2.3.2.1 Floral Diversity

The surveyed sites yielded a total of 256 plant species in 168 genera and 50 families in the sampled areas altogether, of which 90 species were trees and 125 shrubs (Figure 13). The most abundant woody species were *Albizia coriaria, Milicia excelsa, Markhamia lutea, Ficus platyphylla, Mangifera indica, Ficus natalensis, Eucalyptus grandis, Combretum molle, Combretum colinum, Albizia grandibracteata, Vernonia amygdalina, Aeschynomene uniflora, and Artocarpus heterophyllus* in that order.

Non-woody species that spanned most areas included: Cynodon dactylon; Cyperus papyrus, Leersia hexandra, Sparmannia ricincarpa, Sphaeranthus gomphrenoides, Echinochloa pyrimidalis, Typha domingensis, Sporobolus pyrimidalis most of which were typical wetland plants, while some were weeds of cultivation and/or types of wastelands. While shrubs and trees encountered formed the woody population, the majority of the plant species were herbaceous, and these main herbs, graminoids (grasses and grass-like plants), and herbaceous climbers (Appendix 1)

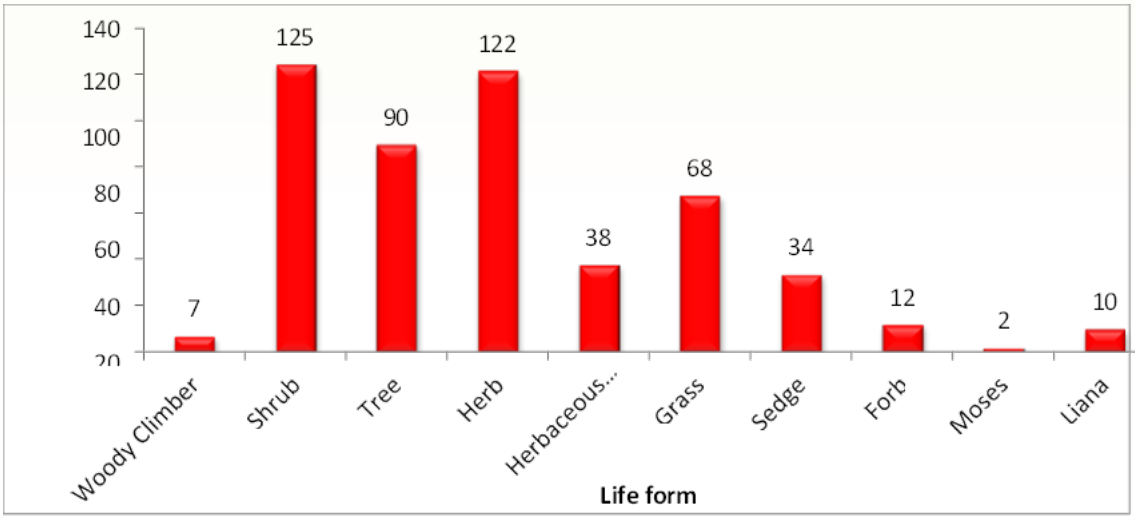


Figure 13: Plant species that were recorded within the project area

Species of conservation concern

Some plant species of conservation concern (in terms of range restriction, rarity, and threat) were observed in the area. These include the following:

a) Species under the IUCN list

Milicia excelsa is one of the species listed under the IUCN Red List of Threatened Species, version 2014.

31 as a species is a nearly threatened species (IUCN red list (2025)

b) Species under NFA reservation

Milicia excelsa and Maesopsis eminii are all on Uganda's National Forestry Authority Reserved Species list and are therefore protected from exploitation and threats to their habitats. Milicia excelsa is on the list due to its commercial logging and timber uses, particularly for high-quality indoor and outdoor furniture, as well as firewood.

c) Invasive species

a. Eichhornia crassipes
b. Mimosa pigra
c. Lantana camara
d. Senna spectabilis
e. Fauna diversity

2.3.2.2 Fauna Diversity

Amphibians

A total of 12 amphibian species belonging to 5 families and 7 genera were recorded in the project area along the waterlogged areas. All these species belong to Order Anura of class Amphibia. The family Hyperolidae had the highest number of species belonging to the genera *Hyperolius, Kassina and Afrixalus.* All identified amphibian species were of Least Concern according to the IUCN Red List of species.

Table 2: Amphibian species in the project area

ORDER	FAMILY	GENUS	SPECIES	COMMON NAME	IUCN STATUS
ANURA	BUFONIDAE	*Amietophrynus*	*gutturalis*	Guttural Toad	(Lc)
ANURA	BUFONIDAE	*Amietophrynus*	*regularis*	Common African toad	(Lc)
ANURA	BUFONIDAE	*Amietophrynus*	*vittatus*	Lake Victoria Toad	(DD)
ANURA	PRYNOBATRACHIDAE	*Phrynobatrachus*	*natalensis*	Natal dwarf puddle frog	(Lc)
ANURA	DICROGLOSSIDAE	*Hoplobatrachus*	*occipitalis*	Crowned bullfrog	(Lc)
ANURA	RANIDAE	*Ptychadena*	*mascareniensis*	Mascarene Grass Frog	(Lc)
ANURA	RANIDAE	*Ptychadena*	*anchietae*	Anchieta's ridged frog	(Lc)
ANURA	HYPEROLIIDAE	*Afrixalus*	*fulvovittatus*	Banded Banana Frog	(Lc)
ANURA	HYPEROLIIDAE	*Hyperolius*	*cinnamomeoventris*	Cinnamon-bellied Reed Frog	(Lc)
ANURA	HYPEROLIIDAE	*Hyperolius*	*kivuensis*	Kivu Reed Frog	(Lc)
ANURA	HYPEROLIIDAE	*Hyperolius*	*viridiflavus*	Common Reed Frog	(Lc)
ANURA	HYPEROLIIDAE	*Kassina*	*senegalensis*	Senegal Kassina	(Lc)

Reptiles

A total of 12 reptile species were recorded in the project area, belonging to 8 families and 9 genera. These species belonged to 4 orders, namely *Serpentes, Squamata, Testudinidae, and Sauria of class Reptilia.* Though IUCN does not evaluate some of these species, those that have been evaluated are of least concern.

Table 3: Reptile species in the project area

No.	ORDER	FAMILY	GENUS	SPECIES	COMMON NAME	IUCN STATUS
1	Sauria	Agamidae	*Agama*	*agama*	Common Agama	LC
2	Sauria	Scincidae	*Trachylepis*	*quinquetaeniata*	Rainbow Skink	NE
3	Sauria	Scincidae	*Trachylepis*	*maculilabris*	Speckle-lipped Skink	NE
4	Sauria	Varanidae	*Varanus*	*niloticus*	Nile Monitor	LC
5	Sauria	Gekkonidae	*Hemidactylus*	*brookii*	Brook's House Gecko	NE
6	Sauria	Gekkonidae	*Lygodactylus*	*gutturalis*	Chevron-throated Gecko	NE
7	Testudinidae	Trionychidae	*Pelomedusa*	*subrufa*	Marsh Terrapin	LC
8	Sauria	Chamaeleonidae	*Chamaeleo*	*gracilis*	Gracile Chameleon	NE
9	Sauria	Scincidae	*Trachylepis*	*striata*	Common Striped Skink	LC
10	Squamata	Colubridae	*Psammophis*	*biseriatus*		LC
11	Serpentes	Elapidae	*Naja*	*melanoleuca*	Forest Cobra	LC
12	Serpentes	Colubridae	*Psammophis*	*sibilans*	Hissing Sand-snake	LC

Mammals

Apart from a few species visually encountered on trees and ground, no concentration of mammals was recorded. A few large mammal species were reported as having previously occurred in the area but have now locally been extirpated. A total of 7 species were reported altogether in the different parts within the project area.

Table 4: Mammal species in the project area

No.	Species Name	IUCN status	
1	*Xerus erythropus* Stripped Ground squirrel	LC	Visually encountered
2	*Cercopithecus aethiops Vervet* Monkey	LC	Visually encountered
3	*Sciurus carolinensis* Eastern Gray Squirrel	LC	Visually encountered
4	*Civetictis Civetta* African Civet	NE	Reported to have existed
5	*Epomops franqueti* Mega Fruit Bat	LC	Visually encountered
6	*Rattus rattus* House Rat	LC	Visually encountered
7	*Lophuromys sikapusi* Rusty-bellied brush-furred rat	LC	Visually encountered

Birds

Although most of the area was cultivated, it still retains a considerable bird population of various categories. A total of 68 bird species were recorded in the project area (Appendix 2), of which 20 were wetland visitors and specialists. None of the birds encountered was a forest specialist. A total of 68 bird species were encountered in the project area, which was an under-representation of the actual avifauna potential of the area. This could be because most bird habitats have been modified, displacing the more specialized forest and wetland species. Birds and other biodiversity have been extensively studied in what is sometimes referred to as the "coffee-banana arc" north of Lake Victoria (Lake Victoria Basin Commission, 2011). The commission's report lists more than 218 species across 26 sites in the basin, including the project area and Mabira Forest.

Table 5: Bird species in the study area

No.	Category	Code	Number of species
1	Forest specialist	FF	0
2	Forest generalist	F	11
3	Tree species	f	14
4	Wetland specialists	W	16
5	Wetland visitor	w	4
6	Grassland specialists	G	5
7	Aerial feeder	Ae	4
8	Palearctic	P	2
9	Afrotropical	A	4
10	Globally Endangered	G-EN	1
11	Endangered	R-EN	1
12	Vulnerable	R-Vu	4
13	Near- threatened	R-NT	2

2.3.2.3 Sensitive ecosystems

The most sensitive ecosystems within the project area are wetlands. These are habitats for birds, mammals, and reptiles. The most extensive wetland is along Lake Victoria (0°14'17.26"N, 32°36'47.79"E) and (0°12'07.21"N, 32°34'19.53"E); however, there are numerous wetlands that seasonally flood during the intensive rainy seasons.

2.3.3 Hydrology

2.3.3.1 Water quality

The physical appearance of the water in the observed wetlands points to heavy pollution and sedimentation. At several sites, the water was described as reddish-brown and dark grey, particularly in Lufuka Wetland, water was dark and emitted foul odors, indicating a high presence of organic waste and possibly sewage infiltration; in Kaliddubi Wetland, a reddish coloration suggests high levels of suspended sediments, likely due to upstream soil erosion and unregulated farming near buffer zones and in Kyetinda Wetland, the site appeared to receive effluents and stormwater discharge, contributing to discolored, turbid water and degraded aquatic life.

Qualitative assessments from key informants support these visual cues and focus group participants, who reported Regular dumping of solid and liquid waste into wetlands, infiltration of domestic sewage and greywater from nearby informal settlements, and industrial runoff and chemical residue, especially near commercial centers and car washing bays.

Furthermore, the presence of algal scum, oily layers, and poor visibility in water bodies suggests nutrient loading (e.g., nitrogen and phosphorus), which can trigger eutrophication and suffocate aquatic organisms. The lack of aquatic plants and declining fish populations in some areas was also noted as a consequence of deteriorating water quality.

2.3.3.2 Streamflow

Another critical indicator of degradation in Makindye Ssabagabo's wetlands is the disruption of natural water flow. Field teams observed that the flow rate was sluggish, with many areas exhibiting stagnant or ponded water; channels were partially or fully blocked by silt, vegetation overgrowth, and improperly disposed debris; and there were signs of artificial diversions and illegal backfilling, which altered natural drainage paths and disconnected wetlands from upstream catchments.

This sluggish flow not only undermines the wetlands' filtration and flood attenuation functions but also creates breeding grounds for disease vectors, increasing the risk of malaria and waterborne illnesses in surrounding communities.



Figure 14:Turbide sluggish stream flow in Lufuka wetland – Mutundwe Ward

Participants in focus group discussions emphasized that the wetlands no longer regulate stormwater effectively, with downstream areas such as Namasuba and Masajja experiencing flash floods during rainy seasons. Additionally, the absence of regular desilting or infrastructure maintenance (e.g., drainage culverts and channels) was highlighted as a major contributing factor to water stagnation.

2.3.4 Socioeconomic status

Wetlands in Makindye Ssabagabo contribute significantly to the municipality's socio-economic fabric by providing natural resources, supporting income-generating activities, and mitigating the effects of environmental hazards such as floods and erosion. However, the socioeconomic value of wetlands is under increasing threat due to urban expansion, poor land use practices, and weak enforcement of environmental regulations.

2.3.4.1 Economic contributions of wetlands

Farming and agriculture: Wetlands support the cultivation of crops such as maize, yams, and sugarcane. Communities rely on fertile wetland soils and moisture for growing food and earning income.

Bricklaying and sand mining: These are major livelihood activities, especially in wetlands in areas like Lufuka, Kaliddubi, and Bunamwaya, where people extract clay and sand for building materials.

Water supply: Wetlands provide water for both domestic and small-scale industrial use, especially in areas not served by piped water infrastructure.

Fishing and aquaculture: Local residents harvest fish such as mudfish and tilapia, though declining biodiversity has made this less viable in recent years.

2.3.4.2 Cultural and social value

Wetlands are not just economic assets; they also hold cultural significance, particularly for communities that use them for traditional ceremonies and gatherings. Wetlands such as Kyentinda and Lufuka provide materials such as papyrus and grass used in roofing huts and making mats, which are cultural artefacts that contribute to community identity and the preservation of heritage, especially within Buganda cultural practices.

2.3.4.3 Health and sanitation impacts

While wetlands contribute to water purification and disease regulation under natural conditions, degraded wetlands now present health risks. Stagnant, polluted water has become a breeding ground for disease vectors like mosquitoes, leading to outbreaks of malaria, bilharzia, and cholera, particularly during the rainy seasons. Uncontrolled solid waste disposal into wetlands has also worsened sanitation conditions in parishes such as Mutungo, Ndejje, and Masajja.

3.3.4.4 Challenges and pressures

The socioeconomic benefits of wetlands are being undermined by

Encroachment and settlement: High population pressure and land scarcity have led to illegal settlements within wetlands, particularly in Lufuka, Kaliddubi, and Namasuba. People are drawn to these areas by the availability of cheap land.

Livelihood dependence vs. degradation: For many, wetlands are a last resort for income—leading to activities like farming and bricklaying that, while sustaining livelihoods, are ecologically destructive.

Perceptions and awareness: Many community members perceive wetlands as wastelands, leading to neglect, degradation, and resistance to restoration initiatives. Communities are ignorant about the value of wetlands.

Conflicting interests: Reports indicate that local council leaders are authorizing illegal land sales within wetlands, while political interference frequently hinders the enforcement of environmental laws.

Lack of a municipal waste dumping site has forced people to dump in wetlands

Inconsistent enforcement/eviction by NEMA

3.3.4.5 Vulnerability and livelihood risks

The degradation of wetlands has had severe consequences for both ecosystems and local livelihoods: Frequent flooding in wetland-adjacent communities in Busabala, Mutungo, and Namasuba has led to loss of property, displacement, and interruption of business and education. Reduced access to clean water and arable land has undermined food security and household income. Decline in biodiversity, including loss of fish and aquatic plants, has further eroded the traditional livelihoods that depend on wetland ecosystems.

2.4 Wetland Restoration

2.4.1 Guidelines for Wetland Restoration

Effective wetland restoration in Makindye Ssabagabo Municipality will require a clear, structured, and context-sensitive approach. The municipality's wetlands have suffered widespread degradation due to rapid urban expansion, illegal encroachment, pollution, and unregulated resource use. Therefore, the guidelines outlined in this plan serve to standardize restoration efforts, ensure community involvement, and align activities with national environmental policies and international best practices.

Prioritize degraded and ecologically significant wetlands

Restoration efforts must begin with a clear identification and prioritization of wetland sites. Priority should be given to: wetlands that have experienced severe degradation but still retain potential for ecological recovery (e.g., Kyetinda, Lubowa, and Namasuba wetlands), wetlands that support flood regulation, biodiversity corridors, and water filtration, and areas where community reliance on wetland services is high and alternatives can be introduced. This step requires mapping, hydrological assessment, and socio-economic analysis to guide targeted interventions.

Restore Hydrological Integrity

Reinstating the natural water flow regime is a core principle of wetland restoration. This includes desilting blocked channels and culverts to restore connectivity between wetlands and upstream/downstream water systems, removing illegal fills and structures that obstruct water flow, and rehabilitating drainage systems where stormwater is redirected into wetlands. Technical assessments must guide hydrological restoration to avoid unintended flooding or erosion in surrounding areas.

Re-establish Native Vegetation

Revegetation is critical for stabilizing soils, enhancing biodiversity, and restoring wetland functions. The following should guide replanting: use of indigenous wetland species such as papyrus, vetiver grass, and aquatic shrubs that are suited to the local ecology, establishment of community-based nurseries to supply planting materials, and engagement of youth and women's groups in planting and aftercare to promote local ownership. Planting should align with the seasonal rainfall calendar for optimal growth and survival.

Control Pollution and Manage Waste

Pollution control is an integral component of restoration. This includes establishing waste collection points and eco-sanitation facilities near vulnerable wetland zones, enforcing laws against dumping solid and liquid waste into wetlands, and creating buffer zones with vegetation that act as filters for runoff and sewage. As part of the waste management efforts, there is a need to acquire land to serve as a dumping site. In addition, community sensitization and coordinated enforcement with NEMA and NWSC are necessary for pollution control to be effective.

Enforce Protective Zoning and Boundaries

Restoration must go hand-in-hand with protection. Boundaries should be demarcated with visible markers and signposts, indicating the 30–50-meter buffer zone as required by law, Gazetted and integrated into municipal physical development plans, monitored by community-based wetland management committees supported by municipal enforcement teams. This zoning must be respected during the planning of infrastructure and human settlement.

Promote Community Participation and Livelihood Alternatives

Wetland restoration will succeed only with active community engagement. Guidelines should include conducting public awareness campaigns on the ecological and economic value of wetlands, supporting alternative livelihoods such as beekeeping, aquaculture, eco-tourism, and sustainable farming, and forming wetland user groups to coordinate community activities and report violations. Social inclusion, especially of women, youth, and the elderly, is essential for sustainable stewardship.

Implement Monitoring and Adaptive Management

Restoration is an ongoing process. Guidelines require the establishment of baseline data on vegetation cover, water quality, and biodiversity, regular monitoring using GIS tools, field assessments, and community reporting, and review of restoration plans based on performance indicators and field feedback. An adaptive approach ensures that restoration efforts remain relevant, responsive, and evidence-based.

2.4.2 Wetland management in Makindye Ssabagabo

Whereas wetlands in Makindye Ssabagabo Municipality serve vital ecological and socio-economic functions, their management remains weak and fragmented, significantly limiting the municipality's ability to protect and sustainably manage these resources. Based on the analysis of field data, interviews, and institutional assessments, several capacity gaps and operational challenges have been identified. The capacity of the municipality to manage wetlands is constrained across several

key areas, including Human Resource Availability, Budgetary Allocation, Infrastructure, Availability of Equipment, Early Warning Systems (EWS), and Governance Structures.

2.4.3 Challenges of Wetland Management and Restoration

The management and restoration of wetlands in Makindye Ssabagabo Municipality, Wakiso District, face several interlinked challenges stemming from socio-economic pressures, governance gaps, and environmental degradation. The challenges are briefly described below.

Encroachment and land use change: One of the major challenges to wetland management in the municipality is the conversion of wetlands into residential areas, farms, and industrial zones. This is aggravated by illegal land sales and encroachment by private developers and individuals. There is also weak enforcement of boundaries and unclear wetland demarcations.

Limited financial resources: The municipality has inadequate funds to effectively implement wetland mapping, protection, restoration, and monitoring. In addition, the municipality relies on external funding with limited long-term sustainability.

Inadequate technical staff: The municipality has limited technical staff to support wetland monitoring, enforcement, and community engagement activities.

Limited community awareness and involvement: There is a low public understanding of wetland ecological benefits. This leads to resistance to wetland restoration from communities that rely on them for farming, bricklaying, or grazing. There is also a lack of effective community engagement in restoration and conservation efforts.

Weak enforcement of environmental laws: There is inadequate implementation of policies by regulatory bodies like NEMA and the Ministry of Water and Environment. In addition, corruption and bribery undermine compliance and accountability. The municipality and local authorities also have a low capacity to monitor and respond to violations.

Institutional overlaps and poor coordination: There are conflicting mandates about wetland management among institutions (e.g., NEMA, WMD, Ministry of Lands, and local government). There is also a lack of a unified wetland management strategy or framework. In addition, there is poor inter-agency communication and collaboration. For example, NEMA sometimes carries out eviction operations without prior consultation or notification to the municipal council, leading to confusion and operational inefficiencies.

Population pressure and urbanization: Rapid urban expansion and population growth increase demand for land and infrastructure. Wetlands are often viewed as idle or "waste" land suitable for settlement or development.

Political interference: Local leaders sometimes support encroachers for political gains. There are also delayed or blocked enforcement actions due to political lobbying or influence. In some instances, some local council leaders (e.g., LC1s) are involved in the sale of wetlands, undermining conservation efforts.

Poor data and information management: There is a lack of accurate and updated maps or ecological assessments. This has fueled disputes over land tenure and unclear ownership of wetland areas.

Pollution and environmental degradation: The municipality is also challenged by the dumping of solid waste and untreated sewage into wetlands. There is no designated dumping site in The Municipality, thus people are forced to dump in the wetlands in Ndejje division. Agricultural runoff in areas of Mutundwe and Seguku that introduces pesticides and fertilizers into water systems further aggravates the problem. Pollution of wetlands from industrial and solid waste, as well as agricultural runoff, leads to flooding, reduced water quality, and the death of aquatic biodiversity.

2.4.4 SWOT analysis of Wetland restoration

Wetland restoration in Makindye Ssabagabo Municipality presents both challenges and opportunities shaped by environmental, institutional, and socio-economic factors. While there are strengths such as established environmental policies and coordination with national agencies like NEMA, weaknesses in enforcement, budget allocation, and institutional capacity hinder effective restoration efforts.

However, there are promising opportunities, including access to global funding, partnerships with NGOs and research institutions, and growing support for nature-based solutions. At the same time, threats such as rapid urban expansion, climate change impacts, land conflicts, and weak policy implementation pose significant risks to wetland conservation (Table.6.).

Table 6: SWORT analysis of Wetland Restoration and Management for Makindye Ssabagabo

Strength	Weakness	Opportunity	Threat
1. Functional Municipal Natural Resources Management Committee. 2. National laws governing wetlands are in place. 3. Existing environmental policies at the municipal level. 4. Good coordination with NEMA, especially in enforcement. 5. Political support for environmental initiatives. 6. Functioning local environment management committees across all the divisions.	1. Inadequate budget allocations for environmental activities. 2. Weak enforcement of wetland regulations. 3. Outdated environmental laws and guidelines. 4. Fragmented or insufficient institutional capacity. 5. Limited data and monitoring mechanisms for wetlands. 6. Limited skilled human resources for wetland management. 7. Inconsistent evictions by NEMA	1) Proximity to the city center and authorities (e.g., KCCA, NEMA) allows for better coordination and oversight. 2) Availability of global and regional funding opportunities (e.g., World Bank) 3) Existing collaboration with research institutions and NGOs. 4) Growing global and national interest in nature-based solutions and wetland restoration. 5) Potential for ecotourism development around wetlands.	1. Rapid urban expansion and land use change are threatening wetland areas. 2. Increased land conflicts and illegal reclamation of wetlands. 3. Rising industrialization and population pressure on wetlands. 4. Weak implementation and political interference in enforcement. 5. Misuse or misallocation of available funding resources. 6. Climate change impacts, including prolonged dry spells and flooding 7. Some individuals hold land titles that were issued before the establishment of NEMA, complicating efforts to enforce current wetland protection regulations.

2.5 Policy and legal frameworks

The wetland management and restoration strategy for Makindye Ssabagabo Municipality is informed and guided by key international and national frameworks that provide legal, institutional, and strategic direction for wetland conservation and sustainable land use. These frameworks will help ensure that local actions are legally grounded, environmentally sound, and aligned with national development priorities.

Table 7: Policy and Legal frameworks

Framework	Relevance to Wetland Management Actions in Ssabagabo Municipality
International frameworks	
Ramsar Convention on Wetlands (1971)	Uganda has been a signatory to the Ramsar Convention on Wetlands since 1988. The Ramsar Convention promotes the wise use and conservation of wetlands, encourages the integration of wetlands into urban planning, and supports community-based wetland restoration.
Convention on Biological Diversity (CBD, 1992)	Uganda ratified the CBD in 1993. The Convention on Biological Diversity encourages the protection of biodiversity in wetland ecosystems, supports participatory conservation planning, and encourages sustainable use of wetland resources.
Paris Agreement (2015)	Uganda ratified the Paris Agreement in 2016, committing to reduce greenhouse gas emissions and enhance resilience through nature-based solutions. This framework supports wetland restoration for flood mitigation, temperature regulation, and carbon mitigation measures.
Sustainable Development Goals (SDGs)	Uganda integrated the SDGs into its national planning frameworks soon after their adoption in 2015. Wetland management aligns particularly with SDG 6 (Clean Water and Sanitation), SDG 13 (Climate Action), and SDG 15 (Life on Land). SDGs reinforce the role of wetlands in sustainable urban development and environmental health.
National frameworks	
Uganda Vision, 2040	Uganda Vision 2040 provides a strong strategic and policy framework for wetland management in the municipality. Its vision of environmental sustainability, urban resilience, and inclusive green growth aligns with local needs to conserve wetlands amidst rapid urban development. Leveraging this national agenda can enhance the effectiveness, legitimacy, and sustainability of wetland management actions in the area.
The National Development Plan IV (NDP IV) (2025/26–2029/30)	The plan provides a strategic and policy framework that directly supports wetland management in Ssabagabo Municipality. It supports restoration, integration of urban planning with environmental sustainability, community engagement, and strengthening local institutions. Implementing wetland management actions in alignment with NDP IV ensures both national coherence and greater access to funding and support.

Framework	Relevance to Wetland Management Actions in Ssabagabo Municipality
Constitution of Uganda (1995)	The 1995 Constitution established the foundational legal principle that wetlands and other natural resources are held by the government in trust for the people. This provision declares wetlands as public trust resources and empowers local governments to ensure the sustainable use and protection of wetlands.
National Environment Act (2019)	Replacing the earlier 1995 Act, the 2019 National Environment Act decentralizes environmental governance, giving local governments like Ssabagabo Municipality more authority to manage and enforce wetland conservation and restoration activities. It allows authorities to issue restoration orders against individuals or entities who encroach or degrade wetlands.
National Wetlands Policy (1995)	This was Uganda's first dedicated policy for wetland protection, developed in response to rapid wetland degradation. It emphasizes the need for wise use, public awareness, and multi-sector coordination. It also supports the sustainable utilization and restoration of wetlands.
Wetlands, Riverbanks & Lakeshores Regulations (2000)	These regulations implement the National Wetlands Policy by defining permissible and restricted activities in wetland areas. They also establish legal protection for wetland ecosystems, guide zoning, buffer management, and restoration processes.
Local Government Act (1997)	Enacted to empower decentralized governance, this Act allows districts and municipalities to manage natural resources, including wetlands. It grants the mandate to local government to manage local wetlands, supports the development of local environmental strategies, and enforcement mechanisms.
Land Act (Cap 227)	This Act recognizes wetlands as protected areas where ownership and use are restricted, prevents the issuance of land titles in wetlands, and helps resolve encroachment issues.
National Climate Change Policy (2015)	Uganda's Climate Change Policy emphasizes ecosystem-based adaptation to climate change. This policy also recognizes wetlands as key ecosystems that buffer communities against climate change impacts, such as floods and droughts. It encourages the conservation and restoration of wetlands as natural carbon sinks that help reduce greenhouse gas concentrations in the atmosphere.

2.6 Institutional arrangement

Wetland management is essential because of its role as a promoter of conservation and wise use of wetlands. The Ministry of Water and Environment, through the Wetlands Management Department, is mandated to protect and conserve wetlands on behalf of Ugandans. The other key partners in the management of wetlands in Uganda include the National Environmental Management Authority, the Ministry of Lands, Housing and Urban Development, District Land Boards and Area Land Committees, District Wetland/Environment Officers, and Lower Local Government Councils (**Table 2**).

Furthermore, as a signatory to the Ramsar Convention, Uganda also has a lean structure as provided for under the convention. This includes the administrative authority, the national focal point, the communication, education, participation, and awareness focal persons for both the government and the NGOs (currently Nature Uganda), and lastly the representative of the scientific and technical panel review. These are responsible for the operationalization of the Ramsar Convention in Uganda.

Table 8: Mandates of key players in Wetland Management

Entity	Roles and responsibilities
Wetlands Management Department	a. Maintaining an up-to-date inventory for wetlands b. Promoting the wise use of wetlands c. Developing wetland policy, guidelines, standards, and legislation; d. Sensitization of the public on the importance of wetlands; e. Restoration and protection of wetlands. f. Technical backstopping and capacity building of local governments g. Monitoring and supervision h. Compliance monitoring and enforcement i. Reviewing Environmental and Social Impact Assessments (ESIAs) and Environmental Audits (EAs) for projects in or around wetlands;
National Environment Management Authority	1. Review and approve all ESIAs countrywide, including projects in wetlands; 2. Issue (and revoke) Wetland resource use permits to applicants who (do not) satisfy conditions laid out in the Wetlands Regulations or specified by NEMA; 3. Conduct periodic inspection and monitoring of wetlands and enforce compliance with the wetland legislation; 4. Ensure the integration of environmental concerns in overall national planning through coordination with the relevant ministries, departments, and agencies of government.
Ministry of Lands, Housing and Urban Development	1. Issue all land titles countrywide and ensure none are issued in wetlands; 2. Cancel titles erroneously issued in wetlands; 3. Develop countrywide base maps indicating land available for titling and that which is not, such as wetlands.

Entity	Roles and responsibilities
Makindye Ssabagabo Municipal Council	a) Conduct due diligence on applications for titles within the municipality to ensure titles are not issued in known wetlands. b) Assist municipality and local environment committees to conserve wetlands within their locality and enforce legal compliance. c) Ensure activities in the catchment area of wetlands do not affect the water level of the wetland. d) Authorize research activities in a protected wetland. e) Formulate bylaws, guidelines, and directives for the management of wetlands. f) With the approval of the environmental officer, order an Environment and Social Impact Assessment (ESIA) for projects that involve the use of a wetland within ten meters of the edge of the wetland.

Source: Adopted from the Office of the Auditor General, 2018 report with changes

Chapter Three: Wetland Restoration Interventions

Introduction

This chapter presents targeted restoration strategies based on clearly defined result areas. The subsequent sections detail specific result areas**,** each defining expected outcomes, key outputs, and actionable strategies. These include interventions for water resources management, service provider engagement, community-led restoration efforts, governance strengthening, livelihood diversification, and ongoing wetland monitoring.

Result-based planning

The development of this plan is based on the results framework concept. A results framework is a planning, communications, and management tool that emphasizes results to provide clarity around your key project objectives. It was introduced in the mid-1990s by USAID as a new approach to monitor its programs throughout the agency. A good results framework is based on a good mix of logic, analysis, standard theories in the technical sector, and the expertise of on-the-ground managers.

The purpose of a results framework is similar to that of a log frame or the theory of change. It helps practitioners to plan and understand the various facets of a project intervention and demonstrate it in a clear graphical manner. A results framework uses a chain of series to display project hypotheses, activities, processes, and the intended changes that are expected to occur throughout the project cycle to achieve the project objectives. It leads to the identification of performance indicators at each stage in this chain, as well as risks that might impede the attainment of the objectives. The elements of the chain of series are laid out under different levels and connected by a cause-and-effect relationship. In short, a person looking at a results framework would be able to understand the basic theory for how key project objectives will be achieved.

Description of Result Areas

3.3.1 Result Area 1: Water Resources Management

Makindye Ssabagabo faces growing water resource challenges stemming from wetland degradation, poor stormwater management, and increased urban runoff. Many wetlands, especially Lubigi, Kyetinda, and Lubowa, are silted and clogged with solid waste and untreated effluents from residential and commercial establishments. The natural hydrology is compromised, contributing to localized flooding and poor water quality. This result area focuses on the rehabilitation of drainage systems, construction of bioswales and artificial wetlands, and controlling pollution sources like illegal waste dumping and open sewerage discharge. These actions are crucial to restoring wetland hydrological functions and improving water retention and filtration capacities.

3.3.2 Result Area 2: Service Providers

There is currently limited capacity and coordination among local tree nurseries and restoration supply chains. Few trained personnel and poor-quality seedlings have constrained replanting efforts. Most restoration activities are still dependent on ad hoc community or NGO-led nurseries with limited technical support and funding. This result area aims at supporting and standardizing nursery operations, particularly for indigenous wetland species, offering training and material support, and linking nursery managers to wetland restoration projects. This will ensure consistent availability of high-quality, ecologically appropriate planting materials for restoration activities.

3.3.3 Result Area 3: Communities (Households & Behavior Change)

Communities living around wetlands in Makindye often rely on them for farming, brickmaking, waste disposal, and other informal uses. However, there is limited awareness of the ecological importance of wetlands and the long-term consequences of degradation. While some community-based efforts exist, overall behavior change remains minimal. This result area, therefore, puts emphasis on community sensitization, formation of wetland stewardship groups, involvement in restoration works, and promotion of eco-friendly livelihood alternatives. Behavior change communication and community capacity building are designed to foster local ownership and sustainable use of wetland resources.

3.3.4 Result Area 4: Wetland Governance (Strengthening Governance Systems)

Governance of wetlands in Makindye Ssabagabo is fragmented, with weak enforcement of environmental laws, overlapping institutional mandates, and insufficient coordination among key actors like MSMC, NEMA, and the Ministry of Lands. Encroachment continues due to delayed or absent policy implementation and enforcement. Proposed actions in this result area focus on developing and enforcing wetland bylaws, establishing clear wetland boundaries, formal gazettement, inter-agency task forces, and empowering local government structures. Strengthening institutional coordination and accountability mechanisms will support long-term wetland protection.

3.3.5 Result Area 5: Livelihood Strengthening

Wetland-dependent communities in Makindye are engaged in unsustainable income activities, such as brick making, farming within wetlands, and/or sand mining. Economic alternatives are scarce, and most households lack access to financial support, skills training, or market linkages for sustainable ventures. The results support eco-enterprise development (e.g., beekeeping, aquaculture, and ecotourism), vocational training, cooperative formation, and access to micro-grants. These interventions are designed to reduce pressure on wetlands by providing viable and eco-friendly sources of income.

3.3.6 Result Area 6: Wetland Monitoring and Assessment

Most wetlands in Makindye Ssabagabo are not officially demarcated or surveyed. This has led to widespread encroachment and difficulty in enforcing protection measures. Baseline data on wetland health, biodiversity, and extent is also lacking or outdated. The result area will include comprehensive wetland surveys, GIS mapping, boundary demarcation using permanent markers, and establishment of community-based monitoring systems. These actions aim to provide a clear legal and spatial basis for planning, restoration, and enforcement efforts.

Chapter Four: Implementation Arrangement

Introduction

This chapter presents the implementation arrangement for the Makindye Ssabagabo Wetland Restoration and Management Plan. It presents the strategic framework through which the plan's six result areas will be operationalized. Each result area is broken down into clearly defined outputs and associated actions, designed to guide the restoration process while addressing key drivers of degradation. The actions are linked to priority wetland locations based on levels of degradation and recovery potential.

Implementation timelines and responsibility

The implementation framework (Table 9) specifies the timeframe for each action, categorized as short-term (0–1 year), medium-term (2–3 years), and long-term (4–5 years), to enable phased execution and monitoring. The framework also identifies the responsible departments and institutions within Makindye Ssabagabo Municipal Council (e.g., the Environment Office, Physical Planning, Engineering, and Law Enforcement units), as well as partner agencies (e.g., NEMA, Ministry of Water and Environment, NGOs, and community structures) tasked with leading or supporting each activity.

Table 9: Wetland restoration implementation Plan

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
Result Area 1: Water resources management	**Output 1: Stormwater drainage systems rehabilitated**	Desilt clogged channels	Namasuba, Ndejje, Seguku	S	Engineering Department of MSMC, Ministry of Works, NWSC
Outcome 1: Wetland hydrological functions are restored and sustained		Reconstruct storm drains to restore functionality of existing systems		S	
		Install trash traps in the upstream channels to prevent solid waste from entering wetland ecosystems.		S	
		Redirect illegal connections to curb unregulated discharges into wetlands		S	
		Train drainage maintenance crews to build capacity for local system upkeep		S	
	Output 2: Pollution sources controlled	Map pollution hotspots	Busabala, Masajja, Mutungo, Ndejje	M	NEMA, NWSC, MSMC
		Install sedimentation ponds to trap pollutants before they enter wetlands		M	
		Promote eco-sanitation by encouraging dry and composting toilets		M	
		Enforce discharge permits to regulate effluent release from industries and homes		M	
		Conduct public sensitization on solid and liquid waste disposal		M	
	Output 3: Wetland water levels regulated	Reopen blocked channels	Busabala, Masajja,	L	MWE, MSMC
		Remove illegal fills to restore natural water pathways		L	
		Construct overflow culverts to manage seasonal water surges		L	

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
		Monitor seasonal flows to inform adaptive wetland management		L	
		Replant hydrophilic vegetation to restore the ecological balance of the wetland systems		L	
Result Area 2: Service providers (e.g., manager of tree nurseries used in restoration)	**Output 1: Wetland plant nurseries supported**	Train tree nursery managers	Mutundwe, Seguku, Ndejje	S	MSMC, NFA, NGOs
Outcome 2: Restoration inputs and technical services are consistently supplied		Supply indigenous seeds (Cyperus, Rotundus & Typha)		S	
		Monitor plant quality to maintain biodiversity integrity		S	
		Set up demonstration sites (restoration) to showcase best practices		S	
		Establish community-based nurseries to engage local ownership		S	
	Output 2: Restoration service provider database created	Identify local providers across the municipality.	Municipal-wide	M	MSMC, Ministry of Trade, MWE
		Develop and maintain a registry/centralized list of actors		M	
		Verify actors' qualifications to ensure service quality		M	
		Publicize vetted providers to enhance visibility		M	
		Link providers to restoration projects to facilitate contracting		M	
	Output 3: Capacity of service providers enhanced	Offer business training	Ndejje, Bunamwaya, Mutungo	L	MSMC, UNDP, Local NGOs
		Provide equipment grants to support operational readiness		L	
		Conduct refresher courses for continuous skills development.		L	
		Facilitate exposure visits for peer learning opportunities		L	
		Develop guidelines/manuals for standardized wetland restoration practices		L	

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
Result Area 3: Communities (households change in behavioral actions)	**Output 1: Community awareness raised**	Conduct radio programs to sensitize communities about environmental responsibility	All Divisions	S	MSMC, Ministry of Education, NGOs
Outcome 3: Wetland-adjacent communities adopt environmentally responsible practices.		Translate IEC materials into local languages to broaden reach		S	
		Organize communal clean-up drives to enhance local responsibility		S	
		Promote school eco-clubs to instill stewardship among the youth		S	
		Use drama & local theatre for culturally resonant messaging.		S	
	Output 2: Wetland user groups established	Mobilize wetland resource users	Masajja, Kyetinda,	M	MSMC, NGOs, CBOs
		Register associations of wetland users for legal recognition		M	
		Train members on wetland laws for legal literacy and empowerment		M	
		Facilitate group meetings to strengthen networks		M	
		Support group savings initiatives for financial inclusiveness.		M	
	Output 3: Behavioral change interventions implemented	Introduce incentives for compliance	Busabala	L	MSMC, NEMA, Local Leadership
		Use community role models to champion responsible behavior.		L	
		Monitor attitudinal shifts to track community response		L	
		Promote eco-certification to recognize sustainable wetland use practices		L	
		Create community bylaws		L	
Result Area 4: Wetland governance	**Output 1: Wetland bylaws developed and enforced**	Draft legal provisions on wetland use	All Divisions	S	MSMC, Ministry of Justice, Municipal Council

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
(strengthening governance systems)					
Outcome 4: Wetland protection is strengthened through policy enforcement and interagency collaboration.		Conduct public consultations to ensure participatory governance		S	
		Translate bylaws into local languages to improve accessibility		S	
		Gazette wetland management bylaws for legal enforcement.		S	
		Train enforcement officers and enforce the bylaws to strengthen compliance		S	
	Output 2: Wetland zones demarcated	Carry out GIS mapping	Mutungo, Namasuba	M	NEMA, MSMC, Ministry of Lands
		Install concrete boundary markers to prevent encroachment		M	
		Publicize zoning regulations to create awareness.		M	
		Remove illegal structures to enforce legal land uses		M	
		Update local development plans to align with conservation needs		M	
	Output 3: Inter-agency collaboration improved	Form a wetland management task force to build a multi-stakeholder approach	Municipal-wide	L	MSMC, NEMA, MWE, Police
		Hold quarterly review meetings		L	
		Develop MoUs		L	
		Share data platforms		L	
		Coordinate enforcement activities		L	
Result Area 5: Livelihood strengthening	**Output 1: Sustainable enterprises supported**	Identify viable eco-businesses	Masajja, Bunamwaya	S	MSMC, Ministry of Gender, NGOs

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
Outcome 5: Eco-friendly income sources are adopted to reduce wetland exploitation.		Promote fish farming in ponded sections of the wetlands for alternative livelihoods.		S	
		Facilitate market access locally and at regional levels		S	
		Support cooperative formation for collective bargaining		S	
		Link to credit institutions to benefit from financial services		S	
	Output 2: Green employment created	Engage the youth in restoration	Mutungo, Ndejje	M	MSMC, Youth Livelihood Fund, UNDP
		Promote eco-tourism to showcase conservation value		M	
		Establish green zones for nature-based enterprises		M	
		Fund wetland ranger jobs for environmental monitoring		M	
		Recognize and reward innovation in wetland conservation to encourage creativity		M	
	Output 3: Skills development offered	Train in value addition	Bunamwaya, Namasuba Masajja, and Ndejje	L	TVET Institutions, MSMC, Enterprise Uganda
		Certify green businesses to enhance market credibility		L	
		Mentor entrepreneurs to develop resilient enterprises		L	
		Establish training hubs to increase access to capacity-building centers		L	
Result Area 6: Wetland monitoring and assessment (demarcation, surveys, etc.)	**Output 1: Wetlands surveyed and mapped**	Conduct GIS mapping of wetland boundaries	All Wetlands	S	NEMA, MSMC, Ministry of Lands
Outcome 6: Wetland health and boundaries		Use drones to collect high-resolution Imagery data		S	

Result Area/Outcomes	Outputs	Actions	Priority Locations	Timeframe (S-Short, M-Medium, L-Long)	Responsible Institutions
are regularly assessed and documented					
		Archive survey data to create a digital repository		S	
		Publish open-access maps to provide information to the stakeholders		S	
		Train local staff in geospatial tools to build in-house skills		S	
	Output 2: Monitoring system established	Install real-time wetland water sensors	Kyetinda & Kaliddubi	M	MWE, MSMC, UNMA
		Set up monitoring dashboards to visualize trends in water level changes		M	
		Create community observer groups to harness citizen science		M	
		Produce quarterly status reports to provide timely updates to stakeholders		M	
		Hold feedback forums with stakeholders to get feedback and facilitate planning		M	
	Output 3: Restoration progress tracked	Create M&E framework to enable municipal-level tracking of restoration efforts	Municipal-wide	L	MSMC, NEMA, Local NGOs
		Track vegetation regrowth as an indicator of wetland restoration success		L	
		Assess compliance indicators to ensure enforcement		L	
		Review field logs for ground-truthing		L	
		Publish impact briefs to disseminate outcomes		L	

4.3 Costing of restoration options

The estimated financial requirements for implementing the Makindye Ssabagabo wetland Restoration and Management Plan are presented in this section (Table 10). The costing covers all proposed interventions across the six strategic result areas.

Each output under these result areas has been assigned a projected annual cost from 2026 to 2030, based on the nature and scale of proposed activities, priority river locations, and logistical needs. The costing is presented in millions of Ugandan Shillings (UGX) and is designed to guide resource mobilization, budgeting, and allocation of funds by the Municipal Council and development partners.

Table 10: Costing of interventions

Output	2026	2027	2028	2029	2030	Total
	Uganda Shillings ('000)					
Stormwater drainage systems rehabilitated	1200	1500	1800	1500	1000	7000
Pollution sources controlled	800	1000	1200	1000	1000	5000
Wetland water levels regulated	700	1000	1300	1500	1500	6000
Wetland plant nurseries supported	500	600	700	600	600	3000
Restoration service provider database created	300	400	500	400	400	2000
Capacity of service providers enhanced	600	700	800	800	800	3700
Community awareness raised	400	500	500	500	500	2400
Wetland user groups established	300	400	500	500	500	2200
Behavioral change interventions implemented	600	700	800	800	800	3700
Wetland bylaws developed and enforced	500	600	600	600	600	2900
Wetland zones demarcated	600	700	800	800	800	3700
Inter-agency collaboration improved	200	300	300	300	300	1400
Sustainable enterprises supported	700	800	900	1000	1000	4400
Green employment created	800	900	1000	1000	1000	4700
Skills development offered	600	700	800	900	900	3900
Wetlands surveyed and mapped	400	500	600	600	600	2700
Monitoring system established	500	600	700	700	700	3200
Restoration progress tracked	300	400	500	500	500	2200

Stakeholder roles and impacts

The key stakeholders include national government agencies, local authorities, community members, private sector players, and civil society organizations (Table 11). Their roles and expected impacts are as follows:

Table 11: Stakeholder roles and impacts

Stakeholder	Role	Expected Impact
Makindye Ssabagabo Municipal Council (MSMC)	Lead implementation, enforce bylaws, and coordinate planning	Improved local wetland governance and enforcement
NEMA	Provide technical guidance, compliance monitoring, and gazettement support	Enhanced national alignment and policy enforcement
Ministry of Water and Environment (MWE)	Support restoration, M&E, and national resource mobilization	Strong technical backstopping and funding leverage
Local Communities	Participate in restoration, protection, and monitoring	Increased ownership and behavior change
CSOs/NGOs (e.g., Nature Uganda)	Awareness, capacity building, and implementation support	Scaled-up community mobilization and livelihood integration
Religious & Cultural Leaders	Promote environmental stewardship messages	Deepened community trust and participation
Private Sector	Support eco-enterprises, CSR, and infrastructure support	Diversified funding and technology input

Coordination arrangements/team roles

A clear coordination structure is required to ensure smooth execution and avoid duplication. The Town Clerk will chair a Municipal Wetland Management Task Force (WMTF). The force will include an environment officer, a physical planner, law enforcement, an NGO representative and community leaders. The role of the force will be to oversee planning, coordination, and review of the implementation of the wetland management and restoration activities.

At the division level, the division environment committees shall monitor restoration activities within parishes and report progress to the WMTF. While at the community level, community wetland monitors shall be trained to support field surveillance and reporting of violations.

Resource Mobilization

Funding will be mobilized from a diverse pool of sources (Table 12)

Table 12: Funding sources

Source	Description
Government allocations	From MSMC and national ministries (MWE, NEMA)
Donor funding	From agencies like UNDP, UNEP, World Bank
NGO partnerships	Co-implementation and co-funding on livelihoods and capacity building
Private sector	CSR investment in green infrastructure, co-sponsorship of restoration sites
Community contributions	In-kind support, such as labor, monitoring, and planting materials

Communication plan

Effective communication will ensure buy-in, learning, and transparency. The communication plan for Makindye Ssabagabo Municipal Council aims to transform awareness into action, build trust with the public, and create a shared vision for the restoration and long-term protection of the municipality's wetland ecosystems.

The communication plan targets five key groups, including communities, local leaders, schoolchildren, civil society organizations, and private sector actors. Each of these groups plays a unique role in either contributing to wetland degradation or supporting its recovery. Communities and youth, for instance, are often at the frontlines—either through unsustainable practices or as agents of change. Local leaders and CSOs have influence, reach, and capacity for mobilization, while private actors, particularly in construction and agriculture, are both stakeholders and potential contributors to restoration financing. To effectively reach these diverse groups, a mix of communication tools and channels is needed, suited to various literacy levels and information access.

Community radio talk shows will be used to hold dialogues in local languages and address myths, concerns, and questions related to wetland protection. Printed IEC (Information, Education, and Communication) materials—such as brochures, posters, and banners in both Luganda and English—will be distributed in public places, schools, and places of worship to reinforce awareness visually.

School campaigns and debate competitions will be used to engage young people, fostering a culture of environmental stewardship from an early age. Religious platforms and local drama groups will serve as culturally relevant avenues to share messages during community gatherings, ensuring that restoration is not just a technical issue but a moral and social responsibility. In addition, social media platforms and WhatsApp groups, particularly those already established among LC leaders, youth groups, and environmental clubs, will be leveraged for rapid, low-cost updates and mobilization.

The communication strategy will focus on a few core messages. First, wetlands are vital life sources—providing water, food, livelihoods, and ecosystem services. Second, the laws and penalties associated with illegal encroachment, dumping, and sand mining will be clearly explained to deter harmful practices. Third, the plan will emphasize the sustainable alternatives available for riverbank users, including eco-enterprises and climate-smart farming. Finally, the strategy will highlight success stories from early adopters—communities and individuals who have embraced

restoration efforts and are seeing benefits. These stories will serve as motivation and proof of concept for others.

4.7 Risk Management

Anticipating and mitigating potential implementation risks will be crucial for the plan implementation.

Table 13: Anticipated Risks and Mitigation

Risk	Mitigation Measure
Encroachment resurgence	Continuous community sensitization and law enforcement
Inadequate funding	Phased implementation, multi-source resource mobilization
Resistance from informal wetland users	Community dialogues and offering of alternative livelihoods
Weak inter-agency coordination	MoUs, joint planning sessions, and reporting protocols
Data gaps or monitoring failure	Capacity building and digital tools (GIS, sensors) integration

Chapter Five: Monitoring, Evaluation and Learning

Introduction

The monitoring, evaluation, and learning framework will ensure continuous tracking, evaluation, and reporting of progress in implementing wetland restoration actions (Table 14). It provides a basis for accountability, learning, and adaptive management. The framework aligns with the restoration goals of Makindye Ssabagabo Municipal Council to restore all existing degraded wetlands, improve livelihoods, and mitigate climate change impacts.

Table 14: M&E framework

Result Area	Output	Indicator	Means of Verification	Baseline	2026	2027	2028	2029	2030
Water Resource Management	Stormwater systems rehabilitated	No. of storm drains desilted	Municipal engineering reports	5	15	25	35	45	60
	Pollution sources controlled	No. of pollution hotspots mitigated	NEMA & NWSC records	2	6	10	15	20	25
	Water levels regulated	% restored hydrological flow	Hydrology monitoring data	20%	30%	45%	60%	75%	90%
Service Providers	Wetland plant nurseries supported	No. of functional nurseries	Inspection reports	3	6	8	10	12	15
	Database created	No. of providers registered	Municipal registry	0	10	20	30	40	50
	Capacity enhanced	No. of trained providers	Training reports	0	15	30	45	60	75
Communities	Community awareness raised	% population aware of wetland roles	Community surveys	25%	40%	55%	70%	80%	90%
	User groups established	No. of active groups	Group registers	2	5	10	15	20	25
	Behavior changes interventions	% adopting eco-friendly practices	Community observations	10%	25%	40%	55%	70%	85%
Governance	Bylaws enforced	No. of bylaws gazetted	Legal documentation	1	2	3	4	5	6
	Zones demarcated	Km of boundary marked	GIS reports	3 km	10	20	30	40	50
	Inter-agency collaboration	No. of taskforce meetings	MoUs & reports	0	4	8	12	16	20
Livelihoods	Sustainable enterprises supported	No. of eco-enterprises supported	MSMC project reports	3	10	20	30	40	50

Result Area	Output	Indicator	Means of Verification	Baseline	2026	2027	2028	2029	2030
	Green jobs created	Number of youth in green employment	Payroll records	10	25	40	55	70	85
	Skills developed	No. trained in green livelihoods	Training certificates	10	30	60	90	120	150
Monitoring & Assessment	Wetlands mapped	No. of wetlands mapped	GIS data	3	6	9	12	15	18
	Monitoring systems installed	No. of stations functional	Sensor dashboards	1	3	5	7	9	10
	Progress tracked	% of outputs tracked in real time	M&E reports	0%	20%	40%	60%	80%	100%

Definition of indicators

Indicators are measurable signs of change that reflect the extent to which planned outputs and outcomes are being achieved over time; thus, they are essential tools for tracking progress, assessing effectiveness, and informing decision-making. Several categories of indicators will be used in the M&E framework of the plan.

Quantitative indicators, for instance, the *number of demarcated wetlands*, *the count of community-based nurseries established*, or *the volume of trash removed from stormwater channels,* are quantifiable outputs that will be tracked over time to demonstrate physical progress. These indicators will particularly be helpful in gauging tangible achievements that are easy to verify through field visits, surveys, and technical reports.

Qualitative indicators, such as the *level of community awareness* about wetland laws or the *perception of wetland importance among local leaders, will* focus on perceptual or descriptive dimensions of change that are not easily counted but are critical to long-term success. These will be assessed through interviews, focus group discussions, and community feedback tools. While less concrete than quantitative data, qualitative indicators provide insights into attitudes, values, and behavior change—factors that underpin sustainable environmental stewardship.

Additionally, **process indicators** such as the *number of training sessions conducted*, *the frequency of stakeholder meetings*, or *the timeliness of monitoring reports will be* used to assess the quality and consistency of implementation. These will help to ensure that activities are being delivered according to the work plan and that institutional processes are functioning effectively.

Finally, **outcome indicators**, for example, *the percentage of wetland area restored*, *the increase in biodiversity*, or *the reduction in flooding events, will* measure the broader results or longer-term impacts of the interventions. These indicators often reflect improvements in environmental, social, or economic conditions. These will typically be monitored annually and will provide the basis for evaluating the success and sustainability of the restoration efforts.

Data collection and analysis

Tools: Monitoring and reporting will employ a combination of tools. These tools include GIS surveys will be used to spatially map and monitor changes in wetland boundaries, vegetation cover, and hydrological patterns.

Community reports will be integral to capturing local observations, experiences, and feedback on wetland use and restoration activities.

Additionally, physical inspections will be regularly conducted to assess on-the-ground progress, including the condition of drainage infrastructure, vegetation regrowth, and the presence of pollution or encroachment.

Stakeholder checklists will be used during review meetings to assess compliance with restoration guidelines, the functionality of institutional frameworks, and the participation of key actors.

Frequency: Output indicators (such as number of channels desilted, wetland boundaries marked, or awareness campaigns conducted) will be assessed quarterly, while impact and outcome indicators (such as improved water quality, biodiversity regeneration, and behavioral change) will be evaluated annually. This phased monitoring will ensure that short-term activities are tracked regularly to inform ongoing improvements, while longer-term changes are assessed in a way that captures trends and sustainability.

Responsible Units: Municipal Environment Department, Physical Planners, NEMA, and partner NGOs. In the M&E, the executive leadership, particularly the mayor and town clerk, will play a critical governance and oversight role to ensure high-level coordination, political support, accountability, and the alignment of restoration efforts with broader municipal development.

Reporting and feedback mechanism

The reporting system for the Makindye Ssabagabo Wetland Restoration and Management Plan is structured around two primary levels.
First, quarterly progress reports will be prepared and submitted by the implementing technical teams to both the Municipal Council and the NEMA. These reports will capture output-level indicators such as the number of wetlands demarcated, trainings conducted, and pollution hotspots managed. They also highlight challenges encountered and recommend corrective actions.

Second, the plan includes annual review forums that bring together a wide array of stakeholders, including local leaders, civil society organizations, technical experts, and community representatives. These forums provide an opportunity to assess outcome indicators such as wetland area restored or changes in community behavior. They serve as a platform for strategic reflection, sharing of lessons learned, and realignment of priorities.

To ensure that progress and decisions are communicated transparently to the public, community updates will be disseminated through local radio programs, municipal notice boards, and mobile van announcements. These methods will enable reaching a wide audience and help build community ownership and awareness around wetland restoration efforts.

Equally important will be the feedback channels designed to capture community perspectives and concerns. These include Local Council I meetings, which provide a grassroots forum for dialogue and reporting. Suggestion boxes, placed at municipal offices and community centers, will allow residents to provide written input anonymously. Additionally, WhatsApp groups for local leaders will offer real-time communication and coordination, facilitating quicker responses to issues on the ground.

Stakeholder engagement (organization structure, capacity building and communication/advocacy)

5.5.1 Organizational Structure

The organizational structure (Figure) for the Makindye Ssabagabo Wetland Restoration and Management Plan M&E Framework depicts a hierarchical yet participatory structure that integrates political leadership, administrative coordination, technical oversight, and community-level engagement.

At the apex of the organogram is the mayor, who provides political oversight and strategic direction for the entire restoration process. The mayor advocates for budgetary support, chairs stakeholder review forums, and ensures that wetland priorities are integrated into broader municipal policies.

Immediately below the mayor is the town clerk, who serves as the **c**hief administrative officer and plays a central role in coordinating departments, aligning restoration activities with municipal plans, and overseeing the preparation and submission of M&E reports. The town clerk also facilitates communication between technical teams and the political leadership.

Reporting to the Town Clerk is the Municipal Coordination Committee (MCC). This is the leading technical body responsible for coordinating day-to-day implementation and M&E activities. The Municipal Environment Officer chairs the MCC and includes key municipal officers such as law enforcement officers, physical planners, engineers, and community development officers. The MCC compiles data, assesses progress against indicators, and liaises with national bodies like NEMA.

Beneath the MCC are the Division Environment Subcommittees, which operate at the division level. These subcommittees are responsible for coordinating activities within their respective divisions, tracking division-specific indicators, and reporting to the MCC. They comprise Environment Focal Persons, LC3 representatives, and technical staff at the division level.

At the grassroots level, the structure includes Community Environment Monitors (CEMs), who are trained volunteers or appointed monitors based at the village or cell level. CEMs play a frontline role in monitoring local wetland activities, reporting violations, tracking environmental changes, and engaging with communities on restoration efforts. They also serve as direct liaisons with the division subcommittees.

5.5.2 Capacity Building

Capacity building will focus on strengthening the knowledge, skills, and institutional effectiveness of all actors involved in wetland protection, from technical officers to community stakeholders. Regular training sessions will be planned for enforcement officers, community champions, and technical staff to ensure they are well-equipped to enforce bylaws, conduct field monitoring, and lead restoration activities. These trainings will cover thematic areas such as environmental law enforcement, ecological restoration techniques, GIS-based monitoring, and community mobilization.

To ensure consistency and scalability, the municipality will oversee the development of standardized training manuals and IEC materials. These materials will be tailored for use in schools, community outreach sessions, and by local leaders. They will include step-by-step restoration guides, wetland management checklists, and educational content that promotes behavioral change. These resources will also support the integration of wetland education into local curricula, further embedding conservation principles into future generations.

5.5.3 Communication & Advocacy

This plan will adopt a multi-channel communication strategy that utilizes locally accessible platforms to reach diverse audiences. These include community radios, school programs, religious institutions, and mobile van campaigns, which will disseminate key messages on wetland values, legal obligations, and community roles. These messages will be delivered in Luganda and other local languages to ensure clarity and cultural relevance.

A key component of the strategy will be the development of visual tools, such as illustrated maps, posters, and signage that raise awareness about wetland boundaries, buffer zones and the benefits of ecosystem services. These tools will be displayed at strategic public points, including trading centers, schools, and municipal offices.

Moreover, the strategy will place strong emphasis on local ownership by mobilizing community champions such as teachers, youth leaders, and respected elders to serve as peer educators and advocates. These champions will lead dialogue sessions, support behavior change campaigns, and serve as role models within their communities.

References

1. Brown, C., Sjögersten, S., Ledger, M. J., Parish, F., & Boyd, D. (2024). Remote Sensing for Restoration Change Monitoring in Tropical Peat Swamp Forests in Malaysia. *Remote Sensing*, *16*(15), 2690.
2. Lake Victoria Basin Commission. (2011). A Study on aquatic biodiversity in the Lake Victoria basin.
3. Mallick, P. H., & Chakraborty, S. K. (2018). Forest, wetland and biodiversity: Revealing multi-faceted ecological services from Eco restoration of a degraded tropical landscape. *Ecohydrology & Hydrobiology*, *18*(3), 278-296.
4. Zhao, Q., Bai, J., Huang, L., Gu, B., Lu, Q., & Gao, Z. (2016). A review of methodologies and success indicators for coastal wetland restoration. *Ecological Indicators, 60*, 442-452.
5. Dennis, T., Adson, T., Edwin, B., Ogwal, F. S., & Richard, K. (2024). Wetland Restoration and Conservation. Case of Uganda. *East African Journal of Environment and Natural Resources*, *7*(1), 391-400.
6. Stapanian, M. A., Adams, J. V., & Gara, B. (2013). Presence of indicator plant species as a predictor of wetland vegetation integrity: a statistical approach. *Plant Ecology*, *214*, 291-302.

Appendices

Appendix 1: List of Plant species that were encountered

No.	Family	*Species*	Authority	Life Form	IUCN status
1	Acanthaceae	*Acanthus polystarchus*	Delile	Shrub	NE
2	Acanthaceae	*Hygrophila uliginosa*	S.Moore	Herb	NE
3	Acanthaceae	*Justicia exigua*	S.Moore	Herb	NE
4	Acanthaceae	*Justicia flava*	(Forssk.) Vahl	Herb	NE
5	Acanthaceae	*Justicia schimperiana*	Hochus. Ex Nees	Herb	NE
6	Acanthaceae	*Pupalia lappacea*	(Linn.) Juss	Herb	NE
7	Acanthaceae	*Thunbergia alata*	Boj ex Sims	Herbaceous climber	NE
8	Alismataceae	*Limnophyton angolense*	Buchen	Herb	NE
9	Amaranthaceae	*Aerva lanata*	(L.) A.L.Juss	Herb	NE
10	Amaranthaceae	*Alternanthera sessilis*	(L.) R.Br. ex DC	Herb	NE
11	Anacardiaceae	*Lannea schweinfurnthii*	Engl	Tree	NE
12	Anacardiaceae	*Mangifera indica*	L	Tree	NE
13	Anacardiaceae	*Ozoroa insignis*	Delile	Shrub	NE
14	Anacardiaceae	*Rhus natalensis*	Bernh ex Krauss	Shrub	NE
15	Anacardiaceae	*Rhus vulgaris*	Meikle	Shrub	NE
16	Apiaceae	*Centella asiatica*	L	Herb	NE
17	Apocynaceae	*Carissa spinarum*	L	woody liana	NE
18	Apocynaceae	*Mondia whitei*	(Hook. F) Skeels	woody Climber	NE
19	Apocynaceae	*Thevetia peruviana*	(Pers.) K. schum	Shrub	NE
20	Araceae	*Pistia stratiotes*	L	Herb	NE
21	Arecaceae	*Borassus aethiopum*	Mart.	Tree	NE
22	Arecaceae	*Phoenix reclinata*	Jacq	Tree	NE
23	Asclepediaceae	*Leptadenia hastata*	Pers.	Herbaceous climber	NE
24	Asparagaceae	*Asparagus africanus*	Lam	Shrub	NE
25	Asparagaceae	*Sansevieria dawei*	Stapf	Herb	NE
26	Asteraceae	*Acmella caulorhiza*	(Dunal) Bitter	Herb	NE
27	Asteraceae	*Adenostemma caffrum*	DC	Herb	NE
28	Asteraceae	*Adenostemma macrophyllum*	(Blume) DC	Herb	NE
29	Asteraceae	*Ageratum conyzoides*	L	Herb	NE
30	Asteraceae	*Aspilia africana*	(Pers.) C.D. Adams	Shrub	NE
31	Asteraceae	*Bidens pilosa*	L	Herb	LC
32	Asteraceae	*Synodrella nodiflora*	Gaertn	Herb	NE
33	Asteraceae	*Conyza floribunda*	Kunth	Herb	NE
34	Asteraceae	*Conyza sumatrensis*	(Retz.) E. H. Walker	Herb	NE
35	Asteraceae	*Enydra fluctuans*	Lour	Herb	NE
36	Asteraceae	*Erlangea marginata*	S.Moore	Shrub	NE
37	Asteraceae	*Laggera alata*	**(D. Don) Sch. Bip.**	Herb	NE
38	Asteraceae	*Pluchea ovalis*	(Pers.) DC	Shrub	NE
39	Asteraceae	*Sphaeranthus gomphrenoides*	O. Hoffm	Herb	NE

40	Asteraceae	*Tithonia diversifolia*	(Hemsl.) A	Herb	NE
41	Asteraceae	*Tridax procumbens*	L	Herb	NE
42	Asteraceae	*Vernonia adoensis*	Schultz	Herb	NE
43	Asteraceae	*Vernonia amygdalina*	Del	Shrub	NE
44	Asteraceae	*Vernonia cinerea*	L	Herb	NE
45	Asteraceae	*Vernonia lasiopus*	). Hoffm	Shrub	NE
46	Azollaceae	*Azolla nilotica*	Decne.dx Mett	Mosses	NE
47	Bignoniaceae	*Jacaranda mimosifolia*	D. don	Tree	NT
48	Bignoniaceae	*Spathodea campanulata*	Beauv.	Tree	NE
49	Bignoniaceae	*Kigelia africana*	(Lam.) Benth.	Tree	NE
50	Bignoniaceae	*Markhamia lutea*	(Denth.) K. Schum	Tree	NE
51	Bignoniaceae	*Stereospermum kunthianum*	Cham	Shrub	NE
52	Boragniaceae	*Cordia monoica*	Roxb	Shrub	NE
53	Boragniaceae	*Trichodesma zeylanicum*	Burm.f.) R.Br.	Herb	NE
54	Burseraceae	*Canarium schweinfurthii*	Engl	Tree	NE
55	Campanulaceae	*Sphenoclea zylenica*	Gartn.	Herb	NE
56	Cannaceae	*Canna indica*	L	Herb	NE
57	Capparaceae	*Capparis erythrocarpos*	Isert	woody liana	NE
58	Capparaceae	*Capparis fascularis*	DC	woody liana	NE
59	Capparaceae	*Capparis tomatoza*	Lam	woody liana	NE
60	Capparaceae	*Crateva adansonii*	L	Shrub	NE
61	Capparaceae	*Maerua angolensis*	DC	Shrub	NE
62	Celastraceae	*Maytenus heterophylla*	Eckl. and Zeyh	Shrub	NE
63	Celastraceae	*Maytenus senegalensis*	(Lam.) Exell	Shrub	NE
64	Combretaceae	*Combretum adenogonium*	Steud. ex A. Rich	Tree	NE
65	Combretaceae	*Combretum collinum*	Fresen	Tree	NE
66	Combretaceae	*Combretum molle*	R.Br ex G. Don.	Tree	NE
67	Combretaceae	*Tarminalia brownii*	Fresen	Tree	NE
68	Combretaceae	*Terminalia schimperiana*	Hochst.	Tree	NE
69	Commelinaceae	*Commelina africana*	L	Forb	NE
70	Commelinaceae	*Commelina beghalensis*	L	Forb	NE
71	Commelinaceae	*Commelina diffuza*	Burm.f.	Forb	NE
72	Convolvulaceae	*Ipomoea acuminata*	Vahl) Roem. & Schult	Herbaceous climber	NE
73	Convolvulaceae	*Ipomoea rubens*	L	Herbaceous climber	NE
74	Convolvulaceae	*Ipomoea wightii*	(Wall.) Choisy	Herbaceous climber	NE
75	Convolvulaceae	*Envolvulus alsinoides*	(L.) Less	Herb	NE
76	Convulvulaceae	*Ipomoea batatas*	(L.) Lam	Herbaceous climber	NE
77	Convulvulaceae	*Ipomoea cairica*	L	Herbaceous climber	NE
78	Convulvulaceae	*Ipomoea sp*		Herbaceous climber	NE

79	Cucurbitaceae	*Mukia maderaspatana*	(L.) M. J. Roem.	Herbaceous climber	NE
80	Cyperaceae	*Bulbostylis ovata*	(Roth) Steven	Sedge	NE
81	Cyperaceae	*Cyperus colymbetes*	kotschy. Peyr	Sedge	NE
82	Cyperaceae	*Cyperus denudatus*	L.f.	Sedge	NE
83	Cyperaceae	*Cyperus difformis*	L	Sedge	NE
84	Cyperaceae	*Cyperus distans*	L. f.	Sedge	NE
85	Cyperaceae	*Cyperus dives*	Delile	Sedge	NE
86	Cyperaceae	*Cyperus latifolious*	Poir	Sedge	NE
87	Cyperaceae	*Cyperus macranthus*	(Boeck.) C.B.Cl	Sedge	NE
88	Cyperaceae	*Cyperus mundtii*	Kunth	Sedge	NE
89	Cyperaceae	*Cyperus papyrus*	L.	Sedge	NE
90	Cyperaceae	*Cyperus rotundus*	L	Sedge	NE
91	Cyperaceae	*Cyperus sp*		Sedge	NE
92	Cyperaceae	*Fimbristylis dichotoma*	(L.) Vahl APNI	Sedge	NE
93	Cyperaceae	*Fuirena pubescens*	Kunth	Sedge	NE
94	Cyperaceae	*Fuirena umbellata*	Rottb	Sedge	NE
95	Cyperaceae	*Kyllinga odorata*	Vahl, Enum	Sedge	NE
96	Euphorbiaceae	*Acalypha bipartita*	Müll. Arg	Shrub	NE
97	Euphorbiaceae	*Acalypha ornata*	Hochst. ex A. Rich	Shrub	NE
98	Euphorbiaceae	*Antidesma membranacea*	Muell. Arg.	Shrub	NE
99	Euphorbiaceae	*Erythrococa bogensis*	Pax	Shrub	NE
100	Euphorbiaceae	*Euphorbia candelabrum*	Trémaux ex Kotschy	Tree	NE
101	Euphorbiaceae	*Euphorbia heterophylla*	L	Herb	NE
102	Euphorbiaceae	*Euphorbia tricalii*	L	Shrub	NE
103	Euphorbiaceae	*Flueggea virosa*	(Willd.) Voigt	Shrub	NE
104	Euphorbiaceae	*Jatropha carcus*	L	Shrub	NE
105	Euphorbiaceae	*Manihot esculenta*	Crantz	Shrub	NE
106	Fabaceae	*Abrus canescens*	L	woody Climber	NE
107	Fabaceae	*Acacia drepanolobium*		Shrub	NE
108	Fabaceae	*Acacia gerrardii*	Benth	Shrub	NE
109	Fabaceae	*Acacia hockii*	De Wild	Shrub	NE
110	Fabaceae	*Acacia polyacantha*	Willd	Tree	NE
111	Fabaceae	*Acacia sieberiana*	DC	Tree	NE
112	Fabaceae	*Aeschynomene schimperi*	A. Rich	Tree	NE
113	Fabaceae	*Aeschynomene uniflora*	E. Mey	Shrub	NE
114	Fabaceae	*Albizia coriaria*	(Welw ex)Oliv	Tree	NE

115	Fabaceae	*Albizia grandibractaeta*	Taub	Tree	NE
116	Fabaceae	*Albizia sp*		Tree	NE
117	Fabaceae	*Allysicarpus vaginalis*		Herb	NE
118	Fabaceae	*Alysicarpus rugosus*	(Willd.) DC	Herb	NE
119	Fabaceae	*Canavaria virosa*	(Wight & Arn.) Gamble.	Herbaceous climber	NE
120	Fabaceae	*Cassia absus*	L	Herb	NE
121	Fabaceae	*Cassia hirsuta*	L	Herb	NE
122	Fabaceae	*Cassia siamea*	Lam	Tree	NE
123	Fabaceae	*Chamaecrista mimosoides*	(L.) Greene	Herb	NE
124	Fabaceae	*Crotalaria spinosa*	Hochst. Ex Benth	Herb	NE
125	Fabaceae	*Desmodium setigerum*	(E. Meyer.) Benth	Shrub	NE
126	Fabaceae	*Desmodium sessilifolium*	(Torr.) Torr. & A. Gray	Herb	NE
127	Fabaceae	*Desmodium triflorum*	(L) DC	Herb	NE
128	Fabaceae	*Dichrostarchys cinerea*	(L.) Wight & Arn)	Shrub	NE
129	Fabaceae	*Indigofera arrecta*	Hochst. ex A. Rich	Shrub	NE
130	Fabaceae	*Indigofera hirsuta*	L	Shrub	NE
131	Fabaceae	*Mimosa pigra*	L	Shrub	NE
132	Fabaceae	*Mimosa pudica*	L	Herb	NE
133	Fabaceae	*Philenoptera laxiflora*	Guill. & Perr.	Tree	NE
134	Fabaceae	*Piliostigma thonningii*	(Schumach.) Milne-Redh	Shrub	NE
135	Fabaceae	*Senna obtusifolia*	L	Shrub	NE
136	Fabaceae	*Senna occidentalis*	L	Shrub	NE
137	Fabaceae	*senna sp*		Shrub	NE
138	Fabaceae	*Senna spectabilis*	(DC.) Irwin & Barneby	Shrub	LC
139	Fabaceae	*Sesbania sesban*	(L.)Merr.	Shrub	NE
140	Fabaceae	*Tamarindus indica*	L	Tree	NE
141	Fabaceae	*Tephrosia lianearis*	(willd.) Pers	Herb	NE

142	Fabaceae	*Teramnus repens*	(Taub.) Baker f.	Herbaceous climber	NE
143	Fabaceae	*Tylosema fassoglensis*	(Schweinf.) Torre & Hillc	Herbaceous climber	NE
144	Fabaceae	*Vigna luteola*	(Jacq.) Benth	Herbaceous climber	NE
145	Lamiaceae	*Bacium sp*		Herb	NE
146	Lamiaceae	*Hoslundia opposita*	Vahl	Shrub	NE
147	Lamiaceae	*Leonotis nepetaefolia*	Willd	Herb	NE
148	Lamiaceae	*Ocimum americanum*	L	Herb	NE
149	Lamiaceae	*Ocimum gratissimum*	L	Shrub	NE
150	Lamiaceae	*Selestemon sp*		Herb	NE
151	Lamiaceae	*Tectona grandis*	L.F	Tree	NE
152	Malvaceae	*Abutiloni s sp*		Shrub	NE
153	Malvaceae	*Hibiscus calyphyllus*	Cav.	Shrub	NE
154	Malvaceae	*Hibiscus cannabinus*	L	Shrub	NE
155	Malvaceae	*Hibiscus ovarifolius*	(Forssk.) Vahl	Shrub	NE
156	Malvaceae	*Sida acuta*	Burn. F	Herb	NE
157	Malvaceae	*Sida ovata*	L	Herb	NE
158	Malvaceae	*Sida rhombifolia*	L	Herb	NE
159	Malvaceae	*Urena lobata*	L	Herb	NE
160	Meliaceae	*Azadirachta indica*	A.Juss.	Tree	NE
161	Meliaceae	*Pseudocedrella kotschyi*	(Schweinf.) Harms	Shrub	NE
162	Meliaceae	*Trichilia emetica*	Vahl	Tree	NE
163	Menispermaceae	*Cissampelos mucronata*	A.Rich.	woody Climber	NE
164	Moraceae	*Artocarpus heterophyllus*	Lam	Tree	NE
165	Moraceae	*Artocarpus heterophylla*		Tree	NE
166	Moraceae	*Ficus ingens*	(Miq.) Miq	Shrub	NE
167	Moraceae	*Ficus exasperata*	Biotik	Tree	NE
168	Moraceae	*Ficus glumosa*	Vahl	Tree	NE

169	Moraceae	*Ficus mucuso*	Welw. Ex Ficalho	Tree	NE
170	Moraceae	*Ficus natalensis*	Hochst	Tree	NE
171	Moraceae	*Ficus ovata*	Vahl	Tree	NE
172	Moraceae	*Ficus sur*	Forssk	Tree	NE
173	Moraceae	*Ficus sp*		Tree	NE
174	Moraceae	*Ficus sycomorus*	L	Shrub	NE
175	Moraceae	*Ficus sp*		Tree	NE
176	Moraceae	*Ficus sur*	Forssk.	Tree	NE
177	Moraceae	*Ficus vallis-choudae*	Delile	Tree	NE
178	Moraceae	*Milicia excelsa*	(Welw.) C.C. Berg	Tree	NE
179	Myrtaceae	*Eucalyptus grandis*	W.Hill ex Maiden	Tree	NE
180	Nymphaeaceae	*Nymphaea lotus*	L	Herb	NE
181	Oleaceae	*Jasminum pauciflorum*	Benth	woody liana	NE
182	Onagraceae	*Ludwigia abyssinica*	A. Rich	Herb	NE
183	Onagraceae	*Ludwigia octovalvis*	Jacq.	Herb	NE
184	Opiliaceae	*Opilia celtidifolia*	(Guill. & Perr.)	woody Climber	NE
185	Papilionaceae	*Erythrina abyssinica*	Lam	Tree	NE
186	Passifloraceae	*Passiflora edulis*	Sims	Herbaceous climber	NE
187	Pedaliaceae	*Sesamum indicum*	L	Herb	NE
188	Phyllanthaceae	*Bridelia miicrantha*	(hochst.) Baill	Shrub	NE
189	Phyllanthaceae	*Phyllanthus amarus*	Schum. & Thonn	Herb	NE
190	Phyllanthaceae	*Phyllanthus ovalifolius*	Forssk.	Shrub	NE
191	Pinaceae	*Pinus caribaea*	Morelet	Tree	NE
192	Poaceae	*Bothrochloa insculpta*	(Hochst ex A.Rich) A. Camus	Grass	NE
193	Poaceae	*Brachiaria comata*	A. Rich	Grass	NE
194	Poaceae	*Brachiaria documbens*	Stapf	Grass	NE
195	Poaceae	*Chloris gayana*	Kunth	Grass	NE

196	Poaceae	*Cynodon dactylon*	(L.) Pers	Grass	NE
197	Poaceae	*Dactyloctenium aegyptium*	(L.) Beauv	Grass	NE
198	Poaceae	*Digitaria abyssinica*	(A. Rich) Stapf	Grass	NE
199	Poaceae	*Digitaria longiflora*	(Retz) Pers	Grass	NE
200	Poaceae	*Digitaria velutina*	(Forssk.) P. Beauv	Grass	NE
201	Poaceae	*Echhinochloa corona*	L	Grass	NE
202	Poaceae	*Echinochloa haploclada*	Stapf	Grass	NE
203	Poaceae	*Echinochloa pyrimidalis*	(Lam.) Hitchc. & Chase.)	Grass	NE
204	Poaceae	*Eragrostis exasperata*	(Thunb.) Steudel	Tree	NE
205	Poaceae	*Eragrostis aspera*	(Jacq.) Nees	Grass	NE
206	Poaceae	*Eriochloa fatmensis*	(Hochst. & Steud.) W.D Clayton	Grass	NE
207	Poaceae	*Hyparrhenia filipendula*	Stapf	Grass	NE
208	Poaceae	*Imperata cylindrica*	(L.) Raeuschel	Grass	NE
209	Poaceae	*Leersia haxandra*	Sw	Grass	NE
210	Poaceae	*Leptochloa filiformis*	P. Beauv	Grass	NE
211	Poaceae	*Melinis repens*	(Willd) Zizka	Grass	NE
212	Poaceae	*Oryza longistaminata*	A.Chev. Roehr	Grass	NE
213	Poaceae	*Oryza sativa*	L	Grass	NE
214	Poaceae	*Panicum maximum*	Jacq	Grass	NE
215	Poaceae	*Paspalum scrobiculatum*	L	Grass	NE
216	Poaceae	*Pennisetum polystachion*	(L.) Schult.	Grass	NE
217	Poaceae	*Pennisetum purpureum*	Schumach	Grass	NE
218	Poaceae	*Phragmites mauritianus*	Kunth	Grass	NE
219	Poaceae	*Saccharum officinarum*	L	Grass	NE
220	Poaceae	*Setaria homonyma*	(Steud.) Chiov	Grass	NE
221	Poaceae	*Setaria sphacelata*	(Schumann) Moss	Grass	NE
222	Poaceae	*Sorghum arundinaceum*	Desv	Grass	NE

223	Poaceae	*Sporobolus festivus*	A. Rich	Grass	NE
224	Poaceae	*Sporobolus pyramidalis*	P. Beauv	Grass	NE
225	Poaceae	*Vossia cuspidata*	(Roxb.) Griff.	Grass	NE
226	Poaceae	*Zee mays*	L.	Grass	NE
227	Polygonaceae	*Persicaria senegalensis*	(Meissn.) Sojak	Forb	NE
228	Polygonaceae	*Persicaria setosula*	(A. Rich.) K. L. Wilson	Forb	NE
229	Polygonaceae	*Rumex bequaertii*	De Wild.	Herb	NE
230	Pontederiaceae	*Eichhornia crassipes*	Mart.	Forb	NE
231	Rhamnaceae	*Maesopsis eminii*	Engl	Tree	NE
232	Rhamnaceae	*Ziziphus pubescens*	Oliv	Shrub	NE
233	Rubiace	*Coffea robusta*	L. Linden	Shrub	NE
234	Rubiace	*Sparmannia ricincarpa*	(Ceckl. Zeyh)	Herb	NE
235	Rubiace	*Spermacocae priorae*		Herb	NE
236	Rubiace	*Spermacoe sp*		Herb	NE
237	Rubiacea	*Gardenia ternifolia*	Schum. Ex Thonn	Shrub	NE
238	Sapindaceae	*Allophylus africana*	P.Beauv	Shrub	NE
239	Sapindaceae	*Cardiospermum sp*	Linn	Herbaceous climber	NE
240	Simaroubaceae	*Harrisonia abyssinica*	Oliv	Shrub	NE
241	Solanaceae	*Capsicum frutescens*	L	Shrub	NE
242	Solanaceae	*Phythalis minima*	L	Herb	NE
243	Solanaceae	*Solanum incanum*	L	Shrub	NE
244	Tiliaceae	*Grewia bicolor*	A. Juss	Shrub	NE
245	Tiliaceae	*Grewia mollis*	A. Juss	Shrub	NE
246	Tiliaceae	*Grewia trichocarpa*	A. Rich	Shrub	NE
247	Tiliaceae	*Triumfetta rhomboidea*	Jacq	Herb	NE
248	Typhaceae	*Typha domingensis*	Pers.	Sedge	NE
249	Verbenaceae	*Lantana camara*	L	Shrub	NE

250	Verbenaceae	*Vitex doniana*	Sweet	Tree	NE
251	Verbenaceae	*Vitex fischeri*	Gurke	Shrub	NE
252	Vitaceae	*Cissus petiolata*	Hook. F.	Herbaceous climber	NE
253	Vitaceae	*Cissus rotundifolia*	(Forssk) Vahl	Herbaceous climber	NE
254	Vitaceae	*Cyphostemma adenocaule*	(Steud.) Descoings	Herbaceous climber	NE
255	Vitaceae	*Cyphostemma cyphopetalum*	(Fresen) Descoings ex Wild & Drumm	Herbaceous climber	NE
256	Zingiberaceae	*Aframomum angustifolium*	(sonn.) K. Schum	Herb	NE

Appendix 2

Appendix 2: List of Bird species that were encountered

ATLAS No.	SPECIES NAME b	Species Category	IUCN status
29	YELLOW-BILLED STORK *Mycteria ibis* 50	RB, W	LC
853	FORK-TAILED DRONGO *Dicrurusadsimilis* 644	RB	LC
39	HADADA *Bostrychia hagedash* 51	W	LC
36	MARABOU STORK *Leptoptiloscrumeniferus* 49	W	LC
30	AFRICAN OPEN-BILLED STORK *Anastomuslamelligerus*43	A,W,G	LC
80	HOODED VULTURE *Necrosyrtesmonachus* 87	RB,-G	EN
76	AFRICAN FISH EAGLE *Haliaeetusvocifer* 137	RB, W	W
75	BLACK KITE *Milvusmigrans* 138	p, A	LC
73	BLACK-SHOULDERED KITE *Elanuscaeruleus* 142	G	LC
125	MARTIAL EAGLE *Polemaetusbellicosus* 134	RB	R-VU
119	AFRICAN HAWK EAGLE *Hieraaetusspilogaster* 128	RB	LC
855	PIED CROW *Corvus albus*654		LC
872	RUPPELL'S LONG-TAILED STARLING *Lamprotornispurpuropter* 1060	F	LC
867	PURPLE STARLING *Lamprotornispurpureus*1059	RB, F	LC
908	BLACK-HEADED WEAVER *Ploceus cucullatus*1165	RB, f	LC
897	SPECTACLED WEAVER *Ploceus ocularis*1177	f	LC
910	YELLOW-BACKED WEAVER *Ploceus melanocephalus* 1173	w	LC
972	BLACK-FACED WAXBILL *Estrildaerythronotus*1228	R(B), F	LC
963	AFRICAN FIREFINCH *Lagonostictarubricata*1239	R(B), G	LC
981	BLACK-AND-WHITE MANNIKIN *Lonchura bicolor*1265	RB	LC
794	COLLARED SUNBIRD *Hedydipna collaris*1080	RB, F	LC
739	PARADISE FLYCATCHER *Terpsiphoneviridis*968	f	LC
508	BLUE SWALLOW *Hirundoatrocaerulea*628	A, w, G,	G-VU,R-VU,

		A e	
520	AFRICAN PIED WAGTAIL *Motacillaaguimp*991	RB, w	LC
363	WHITE-RUMPED SWIFT *Apus caffer*447	RB, Ae	LC
365	LITTLE SWIFT *Apus affinis*443	RB, Ae	LC
358	AFRICAN PALM SWIFT *Cypsiurusparvus*452	RB, Ae	LC
409	ABYSSINIAN GROUND *HORNBILL Bucorvusabyssinicus*527	RB	LC
387	CINNAMON-CHESTED BEE-EATER *Meropsoreobates* 488	RB, F	LC
375	WOODLAND KINGFISHER *Halcyon senegalensis* 475	A	LC
284	AFRICAN MOURNING DOVE *Streptopeliadecipiens* 347	RB	LC
286	RING-NECKED DOVE *Streptopeliacapicola* 346	RB, f	LC
283	RED-EYED DOVE *Streptopeliasemitorquata* 350	f	LC
193	AFRICAN JACANA *Actophilornisafricana* 225	RB, W	LC
213	THREE-BANDED PLOVER *Charadriustricollaris* 239	RB?, W	LC
141	CRESTED GUINEAFOWL *Gutterapucherani*188, 189	RB, F	LC
25	GREY HERON *Ardea cinerea*25	R-NT,W	LC
26	BLACK-HEADED HERON *Ardea melanocephala* 27	w	LC
86	BROWN SNAKE EAGLE *Circaetuscinereus* 98	R(B)	LC
21	LITTLE EGRET Egrettagarzetta 36	RB, W	LC
23	GREAT WHITE EGRET *Egretta alba* 34	W	R-VU
647	WINDING CISTICOLA *Cisticolagalactotes* 860	w	LC
305	EASTERN GREY PLANTAIN EATER *Criniferzonurus* 376		LC
28	HAMERKOP *Scopus umbretta* 42	W	LC
17	CATTLE EGRET *Bubulcus ibis* 32	G	LC
289	LAUGHING DOVE *Streptopelia senegalensis* 351		LC
271	BLUE-SPOTTED WOOD DOVE Turturafer 355	f	LC
562	COMMON BULBUL *Pycnonotus barbatus* 732	f	LC
881	GREY-HEADED SPARROW *Passer griseus*1206	RB	LC
992	PAPYRUS CANARY *Serinuskoliensis* 1287	f	LC
969	COMMON WAXBILL *Estrildaastrild* 1226	f	LC
871	SPLENDID GLOSSY STARLING *Lamprotornissplendidus* 1061	F	LC
796	OLIVE-BELLIED SUNBIRD *Cinnyrischloropygia* 1094	F	LC
881	GREY-HEADED SPARROW *Passer griseus* 1206		LC
576	WHITE-BROWED ROBIN CHAT *Cossyphaheuglini* 751	f	LC
268	AFRICAN GREEN PIGEON *Treroncalva* 358	F	LC
271	BLUE-SPOTTED WOOD DOVE *Turturafer* 355	f	LC
323	WHITE-BROWED COUCAL *Centropussuperciliosus* 406		LC
515	YELLOW WAGTAIL *Motacillaflava* 996	P,w, G	LC
538	LITTLE GREENBUL *Andropadusvirens* 705	F	LC
378	PYGMY KINGFISHER *Ceyxpicta* 478	f,w	
810	COPPER SUNBIRD *Cinnyriscuprea* 1096	f,w	

995	YELLOW-FRONTED CANARY *Serinusmozambicus* 1290		LC
270	TAMBOURINE DOVE *Turturtympanistria* 357	F	LC
456	LESSER HONEYGUIDE *Indicator minor* 566	f	LC

b Names from the Uganda Bird Atlas (Carswell, et al, 2005), with serial numbers, scientific names, and previous (Briton) check-list numbers.

Forest specialist	FF	Endangered	R-EN	Wetland visitor	w	Palearctic	P
Forest generalist	F	Vulnerable	R-Vu	Grassland specialists	G	Afrotropical	A
		Near-				Globally	G-
Tree species	f	threatened	R-NT	Aerial feeder	Ae	Endangered	EN
Wetland specialists	W	Regional responsibility	R-RR				

Appendix 3: Sampled wetlands for fauna

Wetland section	Coordinates	Habitat Description
Natete-Kisigula, Bunamwaya	0°17'52.38"N, 32°31'19.11"E	Farmed wetland
	0°17'14.48"N, 32°31'20.55"E	Degraded wetland
	0°16'57.26"N, 32°31'24.41"E	semi-Natural wetland
	0°16'22.57"N, 32°31'42.59"E	Farmed wetland
	0°14'43.89"N, 32°32'32.22"E	farmed wetland, fish ponds
	0°14'43.89"N, 32°32'32.22"E	Degraded wetland
Nyanama-Zana	0°16'15.10"N, 32°33'34.92"E	Degraded wetland, built up areas
	0°15'53.98"N, 32°33'51.43"E	Degraded wetland, built up areas
Batabata-Kabuma	0°15'34.27"N, 32°34'17.11"E	Degraded wetland, built up areas
	0°15'13.36"N, 32°34'55.31"E	Degraded wetland, built up areas
	0°15'01.73"N, 32°35'51.40"E	Degraded wetland, built up areas
	0°14'17.26"N, 32°36'47.79"E	papyrus wetland
Kajansi-Lweza	0°12'14.33"N, 32°33'08.31"E	papyrus wetland
	0°12'07.21"N, 32°34'19.53"E	lake shoreline

Appendix 2: Wetland inventory

Name: Mayanja Lufuka Seasonal Wetland



Location: 36N 453773 29758

Wetland observation: The wetland is heavily degraded and modified largely by unplanned human settlements, road infrastructure, and subsistence farming from the cultivation of sugarcane and bananas. Scattered native wetland vegetation is comprised mainly of *Cyperus latifolius* and *Colocasia esculenta*.

Ecosystem services: Flood control, Filter waste water and pollutants, Habitat for wetland species.

Biodiversity;

Flora: Dominant wetland vegetation includes *Cyperus latifolius and Pernisetum purpureum*. Other common plant species in degraded sections are *Parthenium spp., Senna spectabilis, Ficus exasperata, Mangifera indica, E. grandis, Ficus bejaminii, and Roystonea regia*.

Fauna: The most common bird species are Cattle egrets, House sparrows, Common bulbul, and Rock pigeons.

Human Encroachment: Human settlements, Road infrastructure, and backfilling using murram.

Restoration interventions: The wetland shares a border with KCCA. Thus, KCCA routinely desilts the Mayaja-Lufuka channel and removal plastic waste.

Water quality:

B.O.D: 0.3mg/l **Temp**: 24.7^{0}C **Soluble Salts**: 0g/l

Name: Mayanja Lufuka Permanent Wetland



Location: 36N 456025 28066

Wetland observation: The wetland is heavily degraded and modified by human activities, including unplanned settlements, subsistence farming for sugarcane, and banana.

Ecosystem services: Flood control, Filter waste water and pollutants, Habitat for diverse species.

Biodiversity;

Flora: Dominant wetland vegetation is *Cyperus papyrus, Cyperus latifolius, and Colocasia esculenta.* Other common plant species in modified sections are the invasive *Parthenium spp, Luffa aegyptiaca, Acanthus ebracteatus, Stachytaropheta jamaicensis, Terminalia catappa, Ricinus cuminis, Lantana camara, Sepium ellipticum, Senna spectabilis, Ficus exasperata, Mangifera indica, E grandis, Ficus bejaminii, and Roystonia regia*.

Fauna: Common bird species are Pied kingfisher, Marabou storks, Hadada ibis, Weavers, Grey heron.

Human Encroachment: Human settlements, Construction, and Illegal waste dumping.

Restoration interventions: The wetland shares a border with KCCA. Thus, KCCA routinely desilts the Mayanja-Lufuka channel and removal plastic waste.

Water quality:

B.O.D: 0.8mg/l **Temp**: 25.5^{0}C **Soluble Salts**: 0g/l

Name: Lufuka Permanent Wetland



Location: 36N 450837 30155

Wetland observation: The wetland is heavily degraded and modified by human activities. The wetland is occupied by Movit industries, brick laying, car washing, illegal waste dumping, unplanned settlements, and subsistence farming, especially piggery.

Ecosystem services: Flood control, wastewater and pollutant filtration, Habitat for diverse species (Birds, amphibians).

Biodiversity

Flora: Dominant wetland vegetation is *Cyperus latifolius* and *Pernisetum purpureum*. Other common plant species in degraded sections include *Parthenium spp., Tithonia rotundifolia, Acanthus ebracteatus, Brugmansia arborea, Senna spectabilis, Ficus exasperata, Mangifera indica, E. grandis, Ficus bejaminii, and Roystonea regia*.

Fauna: The common bird species encountered are the Grey crowned crane, Pied crows, Marabou storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron, weavers (Breeding area),

Human Encroachment: Human settlements (Residential and industrial), illegal waste dumping, car washing, subsistence farming (Maize growing and piggery), and residential waste discharge.

Restoration interventions: No interventions were recorded.

Water quality:

B.O.D: 0.3mg/l **Temp**: 24.7^{0}C **Soluble Salts**: 0g/l

Name: Lufuka Seasonal Wetland



Location: 36N 451404 29243

Wetland observation: This is a heavily degraded and modified wetland largely by dense human settlements. Major industries in the wetland are Sim Plastic Uganda, Movit Industries, and Omega Healing Centre.

Ecosystem services: Flood control, wastewater and pollutant filtration, Habitat for diverse species

Biodiversity;

Flora: Dominant wetland vegetation is *Cyperus latifolius*, *Pernisetum purpureum, with scattered Phoenix reclinata*. Other common plant species in degraded sections include *Parthenium spp., Hyparrhenia filipendula, Senna spectabilis, Ficus exasperata, Mangifera indica, E. grandis, Ficus bejaminii, and Roystonea regia*.

Fauna: The common bird species encountered are the Grey crowned crane, Pied crows, Red wattled lapwing, Cattle egrets, Common bulbul, Red Chested Sunbird, Marabou Storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron, weavers (Breeding area).

Human Encroachment: Human settlements (Residential and industrial), illegal waste dumping, car washing, subsistence farming (maize growing), and residential waste discharge.

Restoration interventions: No interventions are recorded.

Water quality:

B.O.D: 0.4mg/l **Temp**: 30.6^{0}C **Soluble Salts**: 0g/l

Name: Lubigi Channel



Location: 36N 451189 29258

Wetland observation: This is a permanent wetland with sections degraded and modified by dense human settlements and subsistence farmland. Maize crops grown are maize and sugarcane.

Ecosystem services: Flood control, wastewater and pollutant filtration, Habitat for diverse species, and Water for brick laying, Papyrus harvesting.

Biodiversity

Flora: Dominant wetland vegetation is characterised by intact *Cyperus papyrus, C. latifolius,* and *Pernisetum purpureum*. The wetland is also dotted by *Phoenix reclinata*. Other common plant species in degraded sections include *Echinochloa pyramidalis, Cyperus papyrus, Thelypteris acuminata, Paspalum crobiculatum, Parthenium spp., Senna spectabilis, and common bamboo (Bambusa vulgaris).*

Fauna: The common bird species encountered are the Northern grey-headed sparrow, Pied crows, Marabou storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron, Weavers (Breeding area).

Human Encroachment: Human settlements, illegal waste dumping, subsistence farming, and residential waste discharge.

Restoration interventions: No interventions were recorded due to the private ownership of the majority of the wetland.

Water quality:

B.O.D: 0.4mg/l **Temp**: 27.4^{0}C **Soluble Salts**: 0g/l

Name: Kyetinda Permanent Wetland



Location: 36N 449404, 22451

Wetland observation: The dense papyrus wetland has sections that are heavily degraded and modified by human activities. Illegal waste dumping, Small-scale clay and concrete industries, and aquaculture are the main drivers of degradation. Subsistence farmland is dominant, with maize and sugarcane commonly grown.

Ecosystem services: Papyrus harvesting, Aquaculture (Fish ponds), water for tree nurseries, water for clay and concrete industries, Flood control, wastewater and pollutant filtration, Habitat for diverse species, Water for brick laying.

Biodiversity;

Flora: Dominant wetland vegetation is characterised by intact *Cyperus papyrus, C. latifolius,* and *Pernisetum purpureum with some parts infested by water hyacinth*. The wetland is also dotted by *Phoenix reclinata*, Ficus natalensis, F. ovata, Eucalyptus woodlots, and abundant Albizia coraria.

Fauna: The wetland provides a suitable habitat for different species populations, with the majority comprised of Piapiac**,** Cattle Egrets**,** Red wattled lapwing, Northern grey-headed sparrow, Pied crows, Marabou storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron, weavers (Breeding area).

Human Encroachment: Human settlements (Residential and industrial), Construction, illegal waste dumping, car washing, subsistence farming (Maize growing), backfilling using sand, and residential waste discharge.

Restoration interventions: There were no interventions recorded.

Water quality:

B.O.D: 0.8mg/l **Temp**: 22.2^{0}C **Soluble Salts**: 0g/l

Name: Kyetindi seasonal wetland



Location: 36N 452473, 25301

Wetland observation: The wetland is characterised by dense residential and commercial settlements with scattered wetland vegetation. Large sections of the wetland and river channel are degraded, with Parthenium spp more dominant.

Ecosystem services: Flood control, wastewater and pollutant filtration, Habitat for diverse species (Birds, amphibians), Water for brick laying.

Biodiversity;

Flora: Dominant vegetation is characterised by degraded shrubland. The common species recorded were invasive *Parthenium spp, Ricinus cuminis, Roystonia regia, Lantana camara, and Solanum incanum*. Woody species included *Terminalia cattapa, Caliostemon calloptropis, Mangifera indica*, Eucalyptus, and Pinus woodlots.

Fauna: The wetland provides a suitable habitat for different species populations, with the majority comprised of Piapiac**,** Cattle Egrets**,** Red wattled lapwing, Northern grey-headed sparrow, Pied crows, Marabou storks, Ring-necked doves, and Hadada ibis.

Human Encroachment: Car washing, Human settlements (Residential and industrial), and Waste discharge.

Restoration interventions: There were no interventions recorded.

Water quality:

B.O.D: 0.8mg/l **Temp**: 22.2^{0}C **Soluble Salts**: 0g/l

Name: Kaliddubi Permanent Wetland



Location: 36N 454010, 28206

Wetland observation: This is a heavily degraded and modified wetland largely by dense human settlements, with some areas intact. Small-scale industries that manufacture clay stoves and concrete blocks are dominant along the wetland. Subsistence farmland is dominant, with maize and sugarcane commonly grown.

Ecosystem services: The main ecosystem services are Flood control, wastewater and pollutant filtration, and Habitat for birds and amphibians.

Biodiversity;

Flora: Dominant wetland vegetation is characterised by intact *Cyperus papyrus, C. latifolius,* and *Pernisetum purpureum*. The wetland traverses a dominantly Combretum savannah grassland with degraded parts colonised by the invasive Parthenium spp. Other common species include Eucalyptus and Pinus caribaea woodlots, Moringa oleifera, and Mangifera indica.

Fauna: The wetland provides a suitable habitat for different species populations, with the majority comprised of African thrush, Kites, Common bulbul, Speckled mousebird, Piapiac**,** Cattle Egrets**,** Red Wattled lapwing, Northern grey-headed sparrow, Pied crows, Marabou storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron.

Human Encroachment: Human settlements (Residential and industrial), illegal waste dumping, car washing, subsistence farming (Sugarcane, Banana, and Maize growing), and residential waste discharge.

Restoration interventions: Declogging and desilting were done informally on a very small scale. However, Makindye Ssabagabo plans to construct the channel.

Water quality:

B.O.D: 0.7mg/l **Temp**: 24.7^{0}C **Soluble Salts**: 0g/l

Name: Kaliddubi Seasonal Wetland

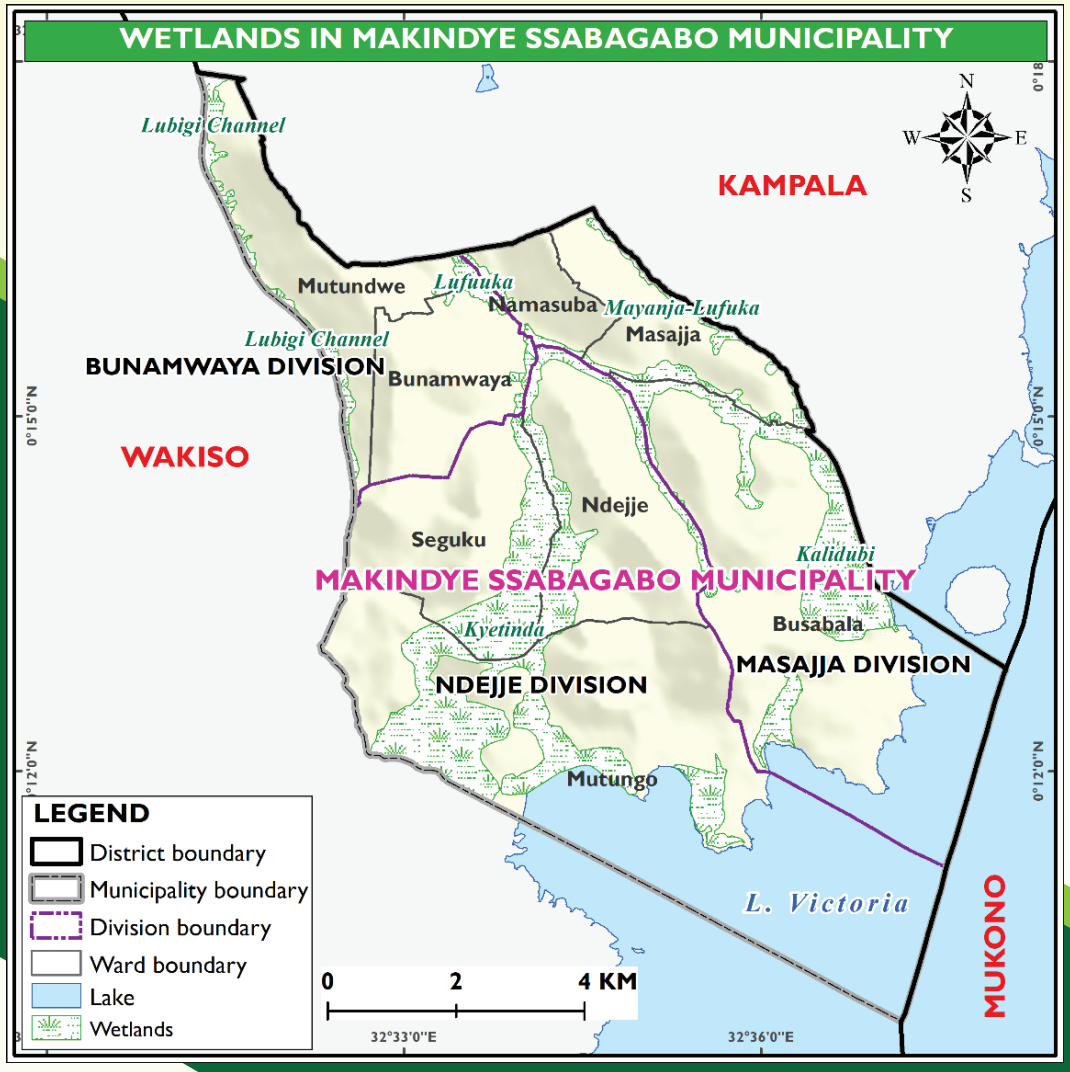


Location: 36N 454010, 28206
Wetland observation: The seasonal wetland is heavily degraded by dense human settlements.
Ecosystem services: Sand mining, Flood control, wastewater and pollutant filtration, Habitat for diverse species, and water for brick laying.
Biodiversity;

Flora: Dominant wetland vegetation is characterized by *C. latifolius and Colocasia esculenta*. The wetland traverses a dominantly Combretum savannah grassland with degraded parts colonised by the invasive Parthenium spp. Other common species include Eucalyptus and Pinus caribaea woodlots, *Artocarpus heterophyllus, Carica papaya, Terminalia superba, Senna spectabilis, Ficus natalensis, Tectona grandis, Pausea americana*

Fauna: The wetland provides a suitable habitat for different species populations, with the majority comprised of Kites, Common bulbul, Speckled mousebird, Piapiac**,** Cattle Egrets**,** Red Wattled lapwing, Northern grey-headed sparrow, Pied crows, Marabou storks, Ring-necked doves, Hadada ibis, Pied kingfisher, Grey heron, weavers (Breeding area).

Human Encroachment: Human settlements (Residential and industrial), illegal waste dumping, car washing, Sand mining, and residential waste discharge.
Restoration interventions: No restoration interventions were recorded.
Water quality:
B.O.D: 0.4mg/l **Temp**: 28.3^{0}C **Soluble Salts**: 0g/l



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THE WORLD BANK